Does It Matter Why Parents Care? Altruism, Warm Glow, and Policy Evaluation

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November 4, 2026

Abstract

Evaluating intergenerational policy requires a model of why parents invest in their children. We show that warm-glow (non-recursive) preferences can reproduce any behavior generated by altruistic (recursive) preferences, but not vice versa. The two nonetheless generally predict different policy responses, and pre-intervention data cannot reveal which is right. A foresight hierarchy, from warm glow to altruism, traces this gap to how many generations of re-optimizing descendants parents anticipate. In an illustrative Becker-Tomes model, altruism and two of many warm-glow specifications fit the same data equally well, yet disagree on whether public education crowds out private spending and whether a universal basic income raises or lowers human capital. We propose choosing policy by minimax regret when the motive is unknown.

Keywords: altruism, warm glow, intergenerational policy analysis

JEL codes: D64, D81, H52, I22, I38, J62.

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1 Introduction

Many of the most consequential policies work across generations. Education subsidies, child tax credits, public health programs, and wealth taxes are meant to help not only today's families but also their descendants. Evaluating such policies requires a model of how parents invest in their children, and any such model has to take a stand on a basic question: why do parents invest at all?

The literature offers two main answers. Under *altruism*, a parent cares about her child's lifetime welfare, which in turn depends on the grandchild's welfare, and so on down the dynasty (Becker, 1974; Barro, 1974; Barro and Becker, 1989). Under *warm glow*, the parent enjoys the act of investing in her child, without regard to what the child later does with it (Andreoni, 1989, 1990). Both formulations are widely used. Quantitative studies typically adopt one of them at the outset, estimate the model, and then run counterfactuals, often with little discussion of the choice. This paper asks whether the choice matters for policy evaluation and, if it does, how a planner should set policy without knowing which motive is right.

We begin with an identification result. Any behavior generated by altruistic parents can also be generated by warm-glow parents. The construction is simple. If the warm-glow parent's payoff from investing equals the discounted value that the altruist attaches to her child's resulting state, the two parents solve the same problem and make the same choices, generation after generation. The warm-glow payoff that achieves this is far from unique, however, because the data reveal it only along the choices that parents actually make.

The reverse does not hold. An altruistic parent values her child through the child's own optimized welfare, so her choices and her child's choices must satisfy an intergenerational Euler equation with a single discount factor. A warm-glow parent pays no attention to how her child will behave, and nothing in the warm-glow model forces her choices to satisfy this equation. Data on several generations can therefore reject altruism, but they can never reject warm glow.

This asymmetry would be of limited interest if the two models agreed about the effects of policy. They generally do not. The disagreement is starkest for policies that change how parental investment turns into child outcomes. A parent who values only her own spending has no reason to respond to such a policy, whereas an altruistic parent does, because the policy changes her child's welfare and, through the child's choices, the welfare of every later generation. If the data do not reject altruism, they cannot tell the two models apart, and the planner faces genuine uncertainty about the model. If the data do reject altruism, only warm glow survives, but its payoff is unrestricted away from observed behavior, so its

counterfactual predictions are not pinned down either. Rejecting altruism therefore makes the identification problem harder rather than easier.

To see where the two models part ways, we introduce a hierarchy indexed by the parent's foresight. A parent with no foresight values her child's state through a fixed function, which is a form of warm glow. A parent with one level of foresight recognizes that her child will re-optimize, a parent with two levels recognizes that her grandchild will do the same, and so on. As foresight increases, the hierarchy converges to altruism, and its value functions do so at a geometric rate. The hierarchy also clarifies the trade-off between the two model classes. Altruism imposes restrictions that make it testable and its counterfactuals determinate. Warm glow can fit any behavior, but only because it leaves counterfactuals open. Intermediate levels of foresight give up some flexibility in exchange for determinacy.

We then ask how large these differences are in a standard setting. Our model of intergenerational human capital investment follows Becker and Tomes (1979), in the simplified form of Solon (2004). We compare altruism with two warm-glow specifications. In the first, parents value their spending on the child; in the second, they value the child's expected income. With logarithmic preferences, the three specifications can be calibrated to be exactly observationally equivalent: they share every technology parameter and generate the same choices and the same joint distribution of parent and child outcomes. Any difference in their predictions therefore reflects the parental motive alone.

The differences turn out to be large. An expansion of public education leaves private spending unchanged under the first warm-glow specification, crowds it out less than one-for-one under the second, and crowds it out more than one-for-one under altruism. A budget-balanced universal basic income (UBI) raises average human capital under the first warm-glow specification and lowers it under the other two. The foresight hierarchy shows that a few generations of foresight close most of the gap between the income-based warm-glow specification and altruism. What separates these two models, then, is whether parents anticipate how their children and grandchildren will respond to a policy, not whether they care about the infinite future.

Because pre-intervention data cannot settle the disagreement, we turn to the choice of policy when the motive is unknown. We use minimax regret, which selects the policy whose worst-case welfare loss across models is smallest, and we measure welfare in each model with that model's own preferences. Altruism has a distinctive property in this exercise. Since each parent's welfare includes that of all her descendants, a policy mistake is counted once for every generation that inherits it. Altruistic welfare is therefore more sensitive to policy, which pulls the robust policy toward the altruistic optimum. In our application, this pull is visible for public education when welfare is measured in utility units. It mostly reflects

how dynastic welfare is normalized, however, and it disappears once losses are expressed in consumption equivalents. For the UBI, the first warm-glow specification implies that redistribution has no efficiency cost, so its optimum sits at the edge of the policy range, and this corner, rather than altruism, determines the robust transfer. For both policies, the motive matters much more for predicted behavior than for the welfare cost of the robust policy, which is small compared with the gains from intervening.

Related Literature

Our paper connects several literatures. The first is the theory of dynastic preferences. Becker (1974) and Barro (1974) introduced altruism toward family members and descendants, Becker and Tomes (1979, 1986) built on it the workhorse model of human capital investment and intergenerational mobility, and Barro and Becker (1989) added endogenous fertility. Non-recursive motives have an equally long history. Andreoni (1989, 1990) formalized warm-glow giving in the context of charity, and Loury (1981) studied intergenerational mobility when parents' preferences over their children are not recursive. We show precisely how the two views relate: they are observationally equivalent in one direction only, and their counterfactual predictions coincide only under a condition that we characterize and that the warm-glow model gives no reason to expect.

The second literature studies decisions under model uncertainty. Hansen and Sargent (2008) analyze policy when the decision maker fears that her model is misspecified. The uncertainty we consider is of a different kind. It concerns preferences rather than the law of motion of aggregate shocks, and the competing models differ qualitatively, recursive versus non-recursive, rather than as small perturbations of a reference model. Our approach is closer to Manski (2004), who develops treatment rules when treatment effects are only partially identified. Counterfactual policy effects have the same structure in our setting, and our use of minimax regret follows his.

Third, the paper speaks to empirical work on how private spending responds to public provision. Peltzman (1973) is an early example, showing that in-kind public provision can crowd out private spending on education. In our setting, holding all pre-intervention data fixed, the crowding-out ratio implied by a structural model ranges from zero to above one depending only on the assumed motive. Model-based predictions of crowding out are therefore only as credible as that assumption, and measured crowding out is, conversely, informative about the motive. A related body of evidence tests altruism directly, typically by asking whether transfers respond to the distribution of income within families, and generally rejects it (Altonji et al., 1997). Our results suggest that such rejections, though informative, make counterfactual analysis harder, because the warm-glow models that survive are unrestricted

away from observed behavior.

Finally, a large quantitative literature evaluates intergenerational policies under a maintained motive. Altruistic dynastic models have been used to study education policy (Abbott et al., 2019), the intergenerational transmission of economic status (Lee and Seshadri, 2019), early childhood development (Daruich, 2026), early and college education and intergenerational persistence (Restuccia and Urrutia, 2004), and neighborhood-based interventions (Chyn and Daruich, 2025). Non-dynastic formulations are common in the analysis of public versus private schooling (Glomm and Ravikumar, 1992), and warm-glow bequest motives are standard in work on wealth inequality (De Nardi, 2004; Cagetti and De Nardi, 2006), estate taxation (Cagetti and De Nardi, 2009), progressive taxation (Morrison, 2024), and the aggregate demand effects of rising inequality (Straub, 2019; Mian et al., 2021). Our results imply that the counterfactual conclusions of this work depend on the maintained motive, and that the calibration targets it typically uses cannot discriminate between motives.

The paper proceeds as follows. Section 2 sets out the environment and the two model classes. Section 3 presents the theoretical results: observational equivalence and its asymmetry (Propositions 1 and 2, illustrated in Example 1), when counterfactual predictions agree (Proposition 3 and Lemma 1), and the depth hierarchy (Proposition 4). Section 4 develops the quantitative application, and Section 5 studies minimax-regret policy, including the amplification of regret under recursion (Proposition 5). Section 6 concludes. Proofs are in Appendix A, additional quantitative results in Appendix B, and numerical details in Appendix C.

2 General Framework

2.1 Environment

Consider a dynasty whose generations are indexed by $t = 0, 1, 2, \ldots$. Generation t inherits a state $x_t \in X \subseteq \mathbb{R}^n$ from its parent. The state collects everything payoff-relevant that passes from parent to child, such as wealth, human capital, or health. Given her state, the parent chooses an action a_t from a feasible set $\Gamma(x_t) \subseteq A \subseteq \mathbb{R}^m$, which covers all decisions with intergenerational consequences, such as spending on education, bequests, and time with children. Her flow payoff $u(x_t, a_t)$ is the utility she obtains within her own lifetime, including any direct cost of investing in the child. Her child inherits the state

$$x_{t+1} = T(x_t, a_t, \varepsilon_t),$$

where ε_t is a shock realized after the parent's choice and drawn independently over time from a fixed distribution. We write $\mathbb{E}$ for the expectation over ε. The deterministic case is the special case of a degenerate shock.

We call the triple (u, T, Γ), together with the distribution of the shock, the economic environment. All the models we consider share it and differ only in the motive behind the parent's choice of a_t. We call a state-action pair (x, a) *interior* if x lies in the interior of X and a in the interior of $\Gamma(x)$. Throughout, we maintain the following conditions.

Assumption 1. (i) $X \subset \mathbb{R}^n$ is compact, and $\Gamma\colon X \rightrightarrows A$ is continuous with nonempty, compact, convex values. For each shock ε, $T(\cdot, \cdot, \varepsilon)$ is continuous and maps $X \times A$ into X.

(ii) The flow payoff u and, for each ε, the transition $T(\cdot, \cdot, \varepsilon)$ are continuously differentiable.

(iii) The altruistic value function V defined in (1) below is continuously differentiable, derivatives of $\mathbb{E}[V(T(x, a, \varepsilon))]$ can be taken inside the expectation, and for each x the function $a \mapsto u(x, a) + \beta\, \mathbb{E}[V(T(x, a, \varepsilon))]$ is strictly concave on $\Gamma(x)$.

Part (i) guarantees that the altruistic problem has a unique bounded continuous value function and that every maximization problem we consider has a solution. Parts (ii) and (iii) make the altruist's choice unique and let us characterize it by a first-order condition whenever it is interior. Part (iii) is a high-level condition on the solution of the altruistic problem, for which sufficient conditions on primitives are well known.[1] We do not assume that choices are interior everywhere. Results that rely on first-order conditions are stated at interior pairs.

2.2 Two Models of Intergenerational Motivation

The altruistic model ($\mathcal{R}$). An altruistic parent values her child's lifetime welfare. Since that welfare depends in turn on how the child treats her own child, the problem is recursive, and the parent's value function satisfies

$$V(x) \;=\; \max_{a \in \Gamma(x)} \Big\{ u(x, a) + \beta\, \mathbb{E}\big[V\big(T(x, a, \varepsilon)\big) \big] \Big\}, \tag{1}$$

where $\beta \in (0, 1)$ measures the strength of altruism. Under Assumption 1(i), this equation has a unique bounded continuous solution (Stokey et al., 1989), and by part (iii) the maximizer, which we denote $\sigma^{\mathcal{R}}(x)$, is unique.

[1] Benveniste and Scheinkman (1979) give conditions under which the value function is differentiable. The application in Section 4 uses logarithmic utility on an unbounded state space and so does not satisfy part (i) literally. Example 1 below verifies the results we need for that environment directly, and the numerical implementation works on a compact grid.

The warm-glow model ($\mathcal{N}$). A warm-glow parent derives utility directly from what she gives her child, rather than from the welfare the child obtains from it. She solves

$$\max_{a \in \Gamma(x)} \big\{ u(x,a) + \Phi(x,a) \big\}, \tag{2}$$

where the function $\Phi \colon X \times A \to \mathbb{R}$ captures her taste for intergenerational transfers. We call Φ *admissible* if it is continuous, continuously differentiable in a, and makes $u(x,\cdot) + \Phi(x,\cdot)$ strictly concave on $\Gamma(x)$ for each x. The warm-glow policy function $\sigma^{\mathcal{N}}$ is then unique and, by Berge's theorem, continuous. The payoff Φ may depend on the child's prospects in any fixed way, for instance through the child's expected state. What makes the problem non-recursive is that Φ does not depend on the child's own choices.

The two models are polar cases. The altruistic parent internalizes the entire future of her dynasty, while the warm-glow parent's concern ends with the act of giving.

3 Theoretical Results

3.1 Observational Equivalence and Testable Restrictions

Suppose the data are generated by altruistic parents. Could an econometrician who observes their choices tell that they are not warm-glow parents? Our first result says that she could not.

Proposition 1 (One-directional observational equivalence)**.** *Let V solve (1) and define $\Phi^*(x,a) = \beta\, \mathbb{E}[V(T(x,a,\varepsilon))]$.*

(a) Φ^* is admissible, and the warm-glow model $\mathcal{N}(\Phi^*)$ generates the same policy function as $\mathcal{R}$. Consequently, the two models produce the same joint distribution of $(x_t, a_t)_{t \geq 0}$ for every initial condition x_0.

(b) Φ^* is not unique. At any state x at which $(x, \sigma^{\mathcal{R}}(x))$ is interior, every admissible $\hat{\Phi}$ with $\nabla_a \hat{\Phi}(x, \sigma^{\mathcal{R}}(x)) = \nabla_a \Phi^*(x, \sigma^{\mathcal{R}}(x))$ generates the same choice. For example, for any $s > 0$, $\Phi_s = (1+s)\Phi^* + s\,u$ generates $\sigma^{\mathcal{R}}$ at every state, yet $\nabla_a \Phi_s \neq \nabla_a \Phi^*$ at every feasible $a \neq \sigma^{\mathcal{R}}(x)$.

The construction in part (a) is the whole argument. With $\Phi = \Phi^*$, the warm-glow parent's objective is exactly the expression the altruist maximizes on the right-hand side of the Bellman equation, so both choose the same action. Since the two models also share the technology, they generate the same data in every respect, including cross-generational statistics such as the intergenerational elasticity of income.

Part (b) says that the matching warm-glow function is far from unique. Choices depend on Φ only through its slope in the action at the chosen action. The data therefore reveal this slope at one action per state, the action parents actually choose, and leave Φ unrestricted at every other action. This freedom will matter when we turn to counterfactuals.

Observational equivalence runs in one direction only. A warm-glow model can always mimic altruism, but altruism generally cannot mimic warm glow. The reason is that altruism ties the choices of successive generations together. Combining the parent's first-order condition with the envelope condition for her child's value gives an intergenerational Euler equation.

Proposition 2 (Asymmetric testability). *(a) Let $n = m = 1$ and $T_a \neq 0$, and suppose the ratio $\chi(x,a) \equiv T_x(x,a,\varepsilon)/T_a(x,a,\varepsilon)$ does not depend on ε, as when shocks enter additively or multiplicatively. Under $\mathcal{R}$, let $a_t = \sigma^{\mathcal{R}}(x_t)$, $x_{t+1} = T(x_t, a_t, \varepsilon_t)$, and $a_{t+1} = \sigma^{\mathcal{R}}(x_{t+1})$. If (x_t, a_t) is interior and (x_{t+1}, a_{t+1}) is interior for almost every realization of ε_t, then*

$$u_a(x_t,a_t) \;=\; -\beta\,\mathbb{E}_t\Big[T_a(x_t,a_t,\varepsilon_t)\,R(x_{t+1},a_{t+1})\Big], \qquad R(x,a) \;\equiv\; u_x(x,a) - u_a(x,a)\,\chi(x,a), \tag{3}$$

where subscripts denote partial derivatives and $\mathbb{E}_t$ is the expectation conditional on x_t.

(b) For any n and m, every continuous policy function σ with $\sigma(x) \in \Gamma(x)$ is generated by some admissible warm-glow function.

The left-hand side of (3) is the parent's marginal cost of investing. The right-hand side is the discounted expected marginal return, evaluated through the child's own behavior. The term $R(x_{t+1}, a_{t+1})$ is the child's marginal value of the state she inherits: her direct flow benefit u_x plus the effect of her state on the grandchild's state. That effect equals the effect of χ additional units of the child's own investment, so the child values it at her own marginal cost of investing, $-u_a\chi$. Given (u, T), every term in (3) is a function of the observed states and choices of two consecutive generations, so the equation is a conditional moment restriction with a single unknown parameter. Under altruism, it must hold at every state with the same β.

A warm-glow parent's first-order condition, $u_a + \Phi_a = 0$, involves only her own state and action. For her choices to satisfy (3), the ratio of her marginal cost to her child's expected marginal return would have to equal the same constant at every state, and nothing in the warm-glow model makes it so. Part (b) states the asymmetry in its strongest form: given the flow payoff, warm glow can reproduce any policy whatsoever, so no data on choices can refute it. The following example shows that the Euler equation, by contrast, has real bite.

Example 1 (Logarithmic utility and Cobb–Douglas technology)**.** Let the state be human capital $h > 0$ and the action the parent's spending a on her child, with

$$u(h, a) = \log(wh - a), \qquad \Gamma(h) = [0, wh), \qquad T(h, a, \varepsilon) = Z\, h^{\rho} a^{\gamma} \varepsilon,$$

where $w, Z > 0$, $\rho \geq 0$, $\gamma > 0$, $\rho + \gamma < 1$, and $\log \varepsilon_t$ is i.i.d. with a finite mean. Write $\lambda = a/(wh)$ for the share of resources spent on the child.

(i) *Altruism.* For every $\beta \in (0, 1)$, the value function is $V(h) = A + B \log h$ with $B = 1/(1 - \beta(\rho + \gamma))$, and the unique optimal policy is a constant share

$$\lambda_{\mathcal{R}}(\beta) \;=\; \frac{\beta\gamma}{1 - \beta\rho}. \tag{4}$$

(ii) *Warm glow.* For a continuous function $b > 0$, the warm-glow payoff $\Phi(h, a) = b(h) \log a$ generates the share $\lambda(h) = b(h)/(1+b(h))$. Every continuous share function with values in $(0, 1)$ arises in this way.

(iii) *Euler equation.* Here $\chi(h, a) = \rho a/(\gamma h)$ does not depend on the shock, and (3) becomes a restriction on the spending shares of parent and child:

$$\frac{\lambda_t}{1 - \lambda_t} \;=\; \beta\, \mathbb{E}_t \left[\frac{\gamma + \rho\, \lambda_{t+1}}{1 - \lambda_{t+1}} \right]. \tag{5}$$

The example makes the asymmetry concrete. As β ranges over $(0, 1)$, the altruistic share (4) rises from zero toward $\gamma/(1 - \rho)$, which is the share that a parent who weighted her descendants as much as herself would approach. Given (u, T), the data are therefore consistent with altruism if and only if the share is constant across states and lies below $\gamma/(1 - \rho)$. In that case, (5) identifies the discount factor as $\beta = \lambda/(\gamma + \rho\lambda)$. A warm-glow parent, by part (ii), can have any share. A constant weight $b \geq \gamma/(1 - \rho - \gamma)$ produces a share that no altruist with $\beta < 1$ would choose, and any weight that varies with h produces a share that varies with income, which no altruist in this environment would choose either. Both reject altruism, while warm glow fits trivially.

The test takes the flow payoff and the technology as known. If u and T must also be estimated, (3) becomes a joint restriction on preferences, technology, and motive, but the asymmetry remains: warm glow can always be made to fit, while altruism cannot.

3.2 Implications for Policy Evaluation

Observational equivalence would be harmless if the models that fit the data also agreed about the effects of policy. For policy evaluation, then, the relevant question is whether the altruistic model and its warm-glow twin make the same predictions once the environment changes. The next result gives an exact answer.

Proposition 3 (When do counterfactual predictions agree?)**.** *Let Φ^* be the warm-glow function of Proposition 1, so that $\mathcal{R}$ and $\mathcal{N}(\Phi^*)$ generate the same pre-intervention policy σ. Suppose the environment changes to $(\tilde{u}, \tilde{T}, \tilde{\Gamma})$, which satisfies Assumption 1 with altruistic value function $\tilde{V}$, while the warm-glow function stays fixed at Φ^* and remains admissible.*

(a) At any state x at which the warm-glow parent's counterfactual choice $\tilde{a} = \tilde{\sigma}^{\mathcal{N}}(x)$ is interior, the altruistic parent makes the same choice if and only if

$$\nabla_a \mathbb{E}\big[\tilde{V}\big(\tilde{T}(x,a,\varepsilon)\big)\big] \;=\; \nabla_a \mathbb{E}\big[V\big(T(x,a,\varepsilon)\big)\big] \qquad \text{at } a = \tilde{a}. \tag{6}$$

(b) If only the technology changes ($\tilde{u} = u$, $\tilde{\Gamma} = \Gamma$), the warm-glow parent does not respond, $\tilde{\sigma}^{\mathcal{N}} = \sigma$, and at any state with $(x, \sigma(x))$ interior the two models agree if and only if (6) holds at $a = \sigma(x)$.

Both sides of (6) measure the marginal value, to an altruist, of investing in the child. The right-hand side is the pre-intervention value, which the warm-glow parent carries unchanged into the new environment through Φ^*. The left-hand side is the post-intervention value. It reflects how the intervention changes the child's welfare and, through the child's choices, the welfare of every later generation. The two models agree exactly when the intervention happens to leave this marginal value unchanged at the relevant action. Nothing in the warm-glow model ties the two sides together, so agreement requires special structure in the environment.

The logic is clearest when the intervention changes only the technology. The warm-glow parent's problem is then unaffected, so she makes the same choice as before. The altruistic parent faces a new Bellman equation and adjusts unless the new technology leaves her marginal value of investing unchanged. When the intervention changes the flow payoff or the feasible set, both parents re-optimize, but they value the child's state differently: the warm-glow parent still uses the pre-intervention value embedded in Φ^*, while the altruistic parent uses $\tilde{V}$.

Example 1 (continued). In the environment of Example 1, $\Phi^*(h,a) = \beta B \gamma \log a$ plus terms that depend only on h. Consider two changes in the technology, holding Φ^* fixed. By

Proposition 3(b), the warm-glow parent keeps the share $\lambda_{\mathcal{R}}(\beta)$ in both cases. First, suppose productivity changes from Z to $\tilde{Z}$. By (4), the altruist's share does not depend on Z, so the two models agree. A neutral change in productivity shifts the altruist's value of every state by a constant and leaves her marginal value of investing unchanged, so (6) holds. Second, suppose the return to investment changes from γ to $\tilde{\gamma} \in (0, 1-\rho)$. The altruist's share becomes $\beta\tilde{\gamma}/(1-\beta\rho)$, and the two models disagree at every state. Interventions that change the marginal return to investment, rather than scaling the child's outcome, are the ones that separate the models.

A natural response is to let the warm-glow payoff depend on the child's prospects, for instance through her expected state, so that the payoff changes when the technology does. This narrows the gap but does not close it. When T changes, a parent with such preferences does adjust, because the mapping from her action to her child's state has changed. She still evaluates that state through a fixed function, however, rather than through her child's own optimization, so she ignores how the child and later descendants will respond. The depth hierarchy of Section 3.3 makes this precise.

Combining Propositions 1 and 3 with the non-uniqueness of the warm-glow function shows that pre-intervention data cannot pin down counterfactuals, whether or not they reject altruism.

Lemma 1 (Non-identification of counterfactuals)**.** *(a) If the observed policy is generated by $\mathcal{R}$ for some $\beta \in (0,1)$, it is also generated by $\mathcal{N}(\Phi^*)$. After an intervention, the two models' choices differ at every state at which the warm-glow choice is interior and condition (6) fails.*

(b) Let Φ be an admissible warm-glow function that generates the observed policy σ and remains admissible after an intervention, with warm-glow functions held fixed. For $c \geq 0$, define

$$\Phi_c(x,a) \;=\; \Phi(x,a) - c\,\|a - \sigma(x)\|^2.$$

Every Φ_c is admissible, generates σ, and remains admissible after the intervention. At any state x at which the counterfactual choice under Φ differs from $\sigma(x)$, the counterfactual choices under Φ_c and $\Phi_{c'}$ differ for all $c \neq c'$ for which both are interior.

In case (a), the researcher faces model uncertainty: the data cannot distinguish altruism from warm glow, and the two models disagree about the policy. In case (b), the data may have settled the question of motive in favor of warm glow because (3) fails, but a different problem takes its place. The penalty $-c\|a-\sigma(x)\|^2$ and its slope vanish at every observed choice, so the data cannot detect it. After an intervention, however, it pulls the counterfactual choice

back toward the old choice $\sigma(x)$. With a scalar action, when $\sigma(x)$ and the prediction under Φ are both interior, the prediction under Φ_c lies between them and approaches $\sigma(x)$ as c grows. The data are therefore consistent with a continuum of counterfactual predictions, ranging from the response implied by Φ toward no response at all.

In practice, the lemma bites for interventions that change the flow payoff. A pure change in the technology leaves every warm-glow parent's choice unchanged when Φ does not depend on the technology. With a scalar action, a pure change in the feasible set moves every specification either nowhere or to the same boundary point.

Rejecting altruism therefore does not help. As long as the altruistic model survives the Euler-equation test, it provides at least one counterfactual disciplined by the Bellman equation. Rejecting it removes this anchor and leaves a surviving class that the data do not constrain away from observed behavior.

3.3 The Depth Hierarchy

The two models make different predictions because they treat the child's response differently. To trace how the gap opens up, we construct a family of models, indexed by a foresight depth $k = 0, 1, 2, \ldots$, that runs from warm glow to altruism. Fix a bounded continuous terminal value function $\phi\colon X \to \mathbb{R}$ and define

$$V_0(x) = \phi(x), \qquad V_k(x) = \max_{a \in \Gamma(x)} \Big\{ u(x,a) + \beta\, \mathbb{E}\big[V_{k-1}\big(T(x,a,\varepsilon)\big)\big] \Big\} \quad (k \geq 1). \tag{7}$$

A parent of depth k chooses $\sigma_k(x) \in \arg\max_{a \in \Gamma(x)} \{u(x,a) + \beta\, \mathbb{E}[V_k(T(x,a,\varepsilon))]\}$. She anticipates k generations of optimizing descendants and values the state beyond them through ϕ. At depth zero, she treats her child's value as fixed, which is the warm-glow model with $\Phi(x,a) = \beta\, \mathbb{E}[\phi(T(x,a,\varepsilon))]$. At depth one, she recognizes that her child will optimize, taking the grandchild's value as fixed, and so on.

Proposition 4 (Convergence of the depth hierarchy)**.** *Under Assumption 1 and for any bounded continuous terminal value ϕ:*

(a) The value functions converge uniformly at a geometric rate: $\|V_k - V\|_\infty \leq \frac{\beta^k}{1-\beta} \|V_1 - V_0\|_\infty$.

(b) The policy functions converge uniformly: $\sup_x \|\sigma_k(x) - \sigma^{\mathcal{R}}(x)\| \to 0$ as $k \to \infty$, for any selection of depth-k maximizers.

(c) For any intervention that changes the environment to $(\tilde{u}, \tilde{T}, \tilde{\Gamma})$ satisfying Assumption 1, the same holds for the counterfactual: $\|\tilde{V}_k - \tilde{V}\|_\infty \leq \frac{\beta^k}{1-\beta} \|\tilde{V}_1 - \tilde{V}_0\|_\infty$ and $\tilde{\sigma}_k \to \tilde{\sigma}^{\mathcal{R}}$ uniformly.

The result follows from the contraction mapping theorem. V_k is the k-th iterate of the Bellman operator starting from ϕ, and V is its unique fixed point. The hierarchy therefore splits the gap between the warm-glow and altruistic counterfactuals into contributions from successive generations of foresight. At depth zero, the prediction coincides with a warm-glow counterfactual, and the bound on the remaining gap in values shrinks by a factor of β with each additional generation of foresight. To measure how much foresight a given intervention requires, we define its foresight content at tolerance $\epsilon > 0$ as

$$\mathcal{F}(\epsilon) \;=\; \min\left\{k \geq 0 \;:\; \frac{\|\tilde{\sigma}_k - \tilde{\sigma}^{\mathcal{R}}\|_\infty}{\|\tilde{\sigma}_0 - \tilde{\sigma}^{\mathcal{R}}\|_\infty} < \epsilon\right\}. \tag{8}$$

If $\mathcal{F}(\epsilon)$ is small, a parent who anticipates the responses of only a few descendants behaves almost like a fully altruistic one. The disagreement between depth zero and altruism then turns on whether parents look beyond their own act of giving, not on the infinite horizon. If $\mathcal{F}(\epsilon)$ is large, the counterfactual depends on foresight extending over many generations.

3.4 Discussion

Falsifiability, flexibility, and determinacy. A class of models of parental motivation can have three desirable properties: it can be falsifiable (some policy function lies outside the class), flexible (every policy function lies inside it), and counterfactually determinate (specifications that generate the same pre-intervention policy also generate the same counterfactual policy after every intervention). The first two exclude each other by definition, so the substantive question is how a class trades them off against the third. The altruistic class $\{\mathcal{R}(\beta)\}$ is falsifiable whenever the Euler equation has bite, as in Example 1. It is also determinate. In the scalar case, if (3) holds at an interior state with $u_a \neq 0$, it pins down β, and the Bellman equation then pins down every counterfactual. It is not flexible. The warm-glow class is flexible by Proposition 2(b), but by Lemma 1(b) it is not determinate. Flexibility does not by itself rule out determinacy, since a class that assigned exactly one specification to each policy function would have both properties. The warm-glow class is indeterminate because it leaves the payoff unrestricted away from observed choices, which is also what makes it flexible. The depth hierarchy lies in between. For any two terminal values ϕ and ψ, the contraction property gives $\|\tilde{\mathcal{T}}^k\phi - \tilde{\mathcal{T}}^k\psi\|_\infty \leq \beta^k\|\phi - \psi\|_\infty$, where $\tilde{\mathcal{T}}$ is the post-intervention Bellman operator. The part of the counterfactual that depends on the arbitrary terminal value therefore shrinks geometrically with depth, as the class converges to altruism.

Post-intervention data. Data from after an intervention can help, within limits. A

natural experiment that changes the technology T while leaving the parent's budget set unchanged separates a warm-glow parent whose Φ ignores T, and whose behavior therefore does not change, from parents with more foresight, who adjust. Three caveats apply. First, such an experiment cannot distinguish warm glow that depends on the child's prospects, which also responds to T, from altruism. Second, evidence from one intervention does not resolve the identification problem for another: the counterfactual for a budget-changing policy remains subject to Lemma 1. Third, experiments that shift the technology while holding budget sets fixed are rare. The public education experiment of Section 4.3.1 is one example.

Impure altruism. An impure altruist (Andreoni, 1989, 1990) maximizes $u(x,a)+\Phi(x,a)+\beta\,\mathbb{E}[V(T(x,a,\varepsilon))]$. Setting $\tilde{\Phi} = \Phi + \beta\,\mathbb{E}[V \circ T]$ turns this into a pure warm-glow problem, so the observational equivalence result applies directly. In the same spirit, a depth-k parent can be read as an altruist over the next k generations with warm-glow preferences beyond them.

4 Quantitative Application

We now ask how large these differences are in a model of intergenerational human capital investment. We deliberately use the simplest environment in which the question can be posed, with a single inherited state (human capital), a single intergenerational action (education spending), and logarithmic preferences. In this environment, the altruistic model and two natural warm-glow specifications can be calibrated to be exactly, rather than approximately, observationally equivalent. Every difference in their counterfactual predictions can then be attributed to the parental motive, and none to differences in fit.

4.1 Model Specification

The model follows Becker and Tomes (1979), in the simplified form without financial bequests used by Solon (2004). Each generation lives for one period. A parent enters adulthood with human capital h_t, which also equals her pre-tax earnings. She pays a proportional tax τ on earnings and receives a lump-sum transfer θ from the government, so her resources are

$$y_t \;=\; (1-\tau)\,h_t \;+\; \theta. \tag{9}$$

She chooses private education spending $e_t \geq 0$ on her child and consumes the rest, $c_t = y_t - e_t$. Parents can neither borrow against their children's future earnings nor leave financial bequests, so education spending is the only link between generations. The child's human

capital is

$$h_{t+1} \;=\; Z \cdot h_t^{\rho} \cdot (\mathcal{E} + e_t)^{\gamma} \cdot \varepsilon_t, \tag{10}$$

where $Z > 0$ is productivity, $\rho \in [0, 1)$ captures spillovers from the parent's human capital, $\mathcal{E} \geq 0$ is public education spending per child, $\gamma \in (0, 1 - \rho)$ governs the returns to total education spending, and $\log \varepsilon_t \sim N\big(-\frac{1}{2}\sigma_\varepsilon^2,\ \sigma_\varepsilon^2\big)$, so that $\mathbb{E}[\varepsilon_t] = 1$. Public and private education are perfect substitutes.

The government uses the earnings tax to finance an exogenous level of spending G, the transfer, and public education. In a stationary equilibrium with distribution μ over human capital, its budget constraint is

$$\tau \int h \, d\mu(h) \;=\; G \;+\; \theta \;+\; \mathcal{E}. \tag{11}$$

In the baseline economy, $\theta = \mathcal{E} = 0$ and G equals baseline revenue. We hold G fixed in all counterfactuals. Combining (9) and (11), average resources in any budget-balanced stationary equilibrium are $\int y \, d\mu = \int h \, d\mu - G - \mathcal{E}$, a property we use repeatedly.

In the notation of Section 2, the state is $x = h$, the action is $a = e$, the flow payoff is $u(h, e) = \log\big((1 - \tau)h + \theta - e\big)$, the transition is $T(h, e, \varepsilon) = Z\, h^{\rho}(\mathcal{E} + e)^{\gamma}\varepsilon$, and the feasible set is $\Gamma(h) = \{e : 0 \leq e < y\}$. With $\theta = \mathcal{E} = 0$, this is the environment of Example 1 with $w = 1 - \tau$. The tax and the transfer enter the flow payoff and the feasible set, whereas public education enters the transition. Our two policy experiments therefore correspond to the two cases of Proposition 3: the UBI to part (a) and public education to part (b).[2]

The three specifications of the parental motive share this environment and differ only in how the parent values her child.

Altruism ($\mathcal{R}$). The parent values the child's lifetime welfare:

$$V(h) \;=\; \max_{e \in \Gamma(h)} \Big\{ \log\big(y - e\big) \;+\; \beta\, \mathbb{E}_\varepsilon[V(h')] \Big\}, \qquad h' = Z\, h^{\rho}(\mathcal{E} + e)^{\gamma}\varepsilon. \tag{12}$$

Warm glow ($\mathcal{N}_{\mathrm{I}}$ and $\mathcal{N}_{\mathrm{II}}$). The parent solves the static problem

$$\max_{e \in \Gamma(h)} \Big\{ \log\big(y - e\big) \;+\; \Phi_j(h, e) \Big\}, \qquad j \in \{\mathrm{I}, \mathrm{II}\}. \tag{13}$$

Both warm-glow payoffs are logarithmic, like the utility of consumption, but they apply to

[2] Logarithmic utility is unbounded and the state space is not compact, so Assumption 1 does not hold literally. Example 1 verifies the baseline results directly, and the numerical implementation restricts h to a compact grid. The constraint $e \geq 0$ may bind once $\mathcal{E} > 0$. Where it does, the first-order conditions below hold as inequalities.

different objects. Under *warm glow I*, the parent values her own spending, as in the pure warm-glow formulation of Andreoni (1989, 1990):

$$\Phi_{\mathrm{I}}(h, e) \;=\; \beta_{\mathrm{I}} \log(e). \tag{14}$$

Under *warm glow II*, she values her child's expected income:

$$\Phi_{\mathrm{II}}(h, e) \;=\; \beta_{\mathrm{II}} \log\Big(\mathbb{E}_{\varepsilon}\big[(1-\tau)h' + \theta\big]\Big) \;=\; \beta_{\mathrm{II}} \log\Big((1-\tau)\, Z\, h^{\rho}(\mathcal{E}+e)^{\gamma} + \theta\Big). \tag{15}$$

The three specifications occupy different positions in the framework of Section 3. Warm glow I is the purest non-recursive model. Its payoff depends on neither the technology nor the policy environment, so the parent does not care how her spending is converted into her child's outcomes. Because this payoff is not a function of the child's state, it lies outside the depth hierarchy. Warm glow II is an instance of the warm glow discussed after Proposition 3, in which the payoff depends on the child's prospects. When there is no transfer, it coincides exactly with the depth-zero model, and otherwise it does so approximately (Section 4.4). A warm-glow II parent responds to anything that changes how her spending maps into her child's income, including public education, taxes, and transfers, but she values that income through a fixed function rather than through her child's own choices. The altruistic parent values her child's human capital through V, which builds in the child's optimal response to the policy environment and, through it, the responses of all later generations. Comparing warm glow I with warm glow II therefore isolates the lack of discipline on how the warm-glow function depends on the policy environment, and comparing warm glow II with altruism isolates the role of recursion.

4.2 Baseline Equivalence and Calibration

Without public education or transfers, all three specifications have closed-form solutions, and they can be made to coincide exactly.

Lemma 2 (Baseline equivalence in the application)**.** *Suppose $\theta = \mathcal{E} = 0$. Under each specification, the parent spends a constant fraction of her resources on education, $e = \lambda_m\, y$, with*

$$\frac{\lambda_{\mathcal{R}}}{1-\lambda_{\mathcal{R}}} = \frac{\beta\gamma}{1-\beta(\rho+\gamma)}, \qquad \frac{\lambda_{\mathrm{I}}}{1-\lambda_{\mathrm{I}}} = \beta_{\mathrm{I}}, \qquad \frac{\lambda_{\mathrm{II}}}{1-\lambda_{\mathrm{II}}} = \beta_{\mathrm{II}}\,\gamma. \tag{16}$$

If the preference weights satisfy

$$\beta_{\mathrm{II}} = \frac{\beta}{1-\beta(\rho+\gamma)}, \qquad \beta_{\mathrm{I}} = \gamma\,\beta_{\mathrm{II}}, \tag{17}$$

then the three specifications generate the same policy function and the same law of motion,

$$\log h_{t+1} \;=\; \log Z + \gamma \log\big(\lambda(1-\tau)\big) + (\rho+\gamma)\log h_t + \log \varepsilon_t, \tag{18}$$

and hence the same joint distribution of $(h_t, e_t)_{t\geq 0}$. *The stationary distribution is log-normal, and the mean* μ_h *and variance* σ_h^2 *of* $\log h$ *are*

$$\sigma_h^2 = \frac{\sigma_\varepsilon^2}{1-(\rho+\gamma)^2}, \qquad \mu_h = \frac{\log Z + \gamma \log\big(\lambda(1-\tau)\big) - \sigma_\varepsilon^2/2}{1-\rho-\gamma}. \tag{19}$$

Lemma 2 is a special case of Proposition 1 in which the matching warm-glow function can be written down explicitly. By Example 1, the altruistic value function is $V(h) = A + B\log h$ with $B = 1/(1-\beta(\rho+\gamma))$, so the warm-glow function of Proposition 1 is $\Phi^*(h,e) = \beta B\,\gamma \log e$ plus terms in h alone. Under (17), both Φ_{I} and Φ_{II} equal Φ^* up to terms that do not affect choices, and they do so at every (h,e) in the baseline environment, not only along observed choices. In particular, warm glow I is exactly the warm-glow twin $\mathcal{N}(\Phi^*)$ of Proposition 1, held fixed across policy environments. What the baseline data cannot reveal is how the warm-glow payoff depends on the policy environment. Under altruism, the Bellman equation dictates this dependence, because $\Phi^* = \beta\,\mathbb{E}[V(T(h,e,\varepsilon))]$ changes whenever V or T changes. Under warm glow, it is a modeling choice: warm glow I specifies no dependence at all, and warm glow II specifies dependence only through the child's expected income.

Moments and calibration. Since income $(1-\tau)h$ is a monotone transformation of human capital, the stationary distribution delivers the targeted moments in closed form:

$$\mathbb{E}[h] = e^{\mu_h + \sigma_h^2/2}, \qquad \mathrm{CV} = \sqrt{e^{\sigma_h^2}-1}, \qquad \mathbb{E}[e/y] = \lambda, \qquad \text{rank-rank slope} = \tfrac{6}{\pi}\arcsin\!\Big(\tfrac{\rho+\gamma}{2}\Big). \tag{20}$$

The last expression is the Spearman rank correlation of a stationary bivariate normal vector with correlation $\rho+\gamma$.

We fix the tax rate at $\tau = 0.25$ and the spillover at $\rho = 0.24$, and target four moments: mean human capital $\mathbb{E}[h] = 1$ (a normalization), an average education spending share $\mathbb{E}[e/y] = 0.10$, a coefficient of variation of earnings of 1, and a rank-rank slope of 0.35, close to the U.S. estimate of Chetty et al. (2014). The education share and earnings dispersion are illustrative round numbers. The closed-form calibration proceeds one parameter at a time. The rank-rank slope gives the persistence $\rho+\gamma = 2\sin(0.35\pi/6) = 0.364$, and hence $\gamma = 0.124$. The coefficient of variation gives $\sigma_h^2 = \log 2$, and hence $\sigma_\varepsilon = \sqrt{(1-(\rho+\gamma)^2)\log 2} = 0.775$. The education share gives $\lambda = 0.10$. Mean human capital then gives $Z = 1.496$ through (19), and Example 1 gives $\beta = \lambda/(\gamma+\rho\lambda) = 0.674$. Because

Table 1: Baseline Calibration

Parameter		Value	Target
Fixed			
τ	Tax rate	0.25	Fixed
ρ	Parental spillover	0.24	Fixed
Common to all three specifications			
γ	Education returns	0.124	Rank-rank slope = 0.35
σ_ε	Shock std. dev.	0.775	CV of earnings = 1
Z	HC technology	1.494	$\mathbb{E}[h] = 1$
Specification-specific preference weight			
β	Altruism ($\mathcal{R}$)	0.675	$\mathbb{E}[e/y] = 0.10$
β_{I}	Warm glow I ($\mathcal{N}_{\mathrm{I}}$)	0.111	$\mathbb{E}[e/y] = 0.10$
β_{II}	Warm glow II ($\mathcal{N}_{\mathrm{II}}$)	0.896	$\mathbb{E}[e/y] = 0.10$
Model moments (identical across specifications)			
Mean human capital		1.00	1
Education share		0.100	0.10
CV of earnings		1.00	1
Rank-rank slope		0.350	0.35
IGE of earnings		0.364	Untargeted

Note: The closed form of Lemma 2 gives $\gamma = 0.124$, $\sigma_\varepsilon = 0.775$, $Z = 1.496$, $\beta = 0.674$, $\beta_{\mathrm{I}} = 0.111$, and $\beta_{\mathrm{II}} = 0.893$. Reported values match the targets on the discretized state space (Appendix C). Baseline policy: $\theta = \mathcal{E} = 0$, with government spending G equal to baseline revenue. Because the three specifications generate the same policy function and law of motion, every pre-intervention moment, targeted or not, is identical across them.

the numerical solution uses a discretized state space, we refine these values by matching the four moments computed on the grid (Appendix C). The refinement changes Z to 1.494 and β to 0.675 and leaves γ and σ_ε unchanged to three decimals. The warm-glow weights then follow from (17). Table 1 reports the values used in all experiments.

The calibration has three properties that matter for what follows. First, the three specifications agree on every pre-intervention statistic, including untargeted ones such as the intergenerational elasticity of earnings (Solon, 1999) and the full joint distribution of parent and child outcomes. By Example 1, a constant education share is consistent with altruism if and only if it lies below $\gamma/(1-\rho) \approx 0.164$. The targeted share of 0.10 satisfies this bound, so the data satisfy the Euler equation exactly, and we are in case (a) of Lemma 1: altruism cannot be rejected, and no pre-intervention evidence can favor one specification over another. By contrast, a warm-glow-I parent with a weight β_{I} above $\gamma/(1-\rho-\gamma) \approx 0.196$ would spend a share that no altruist would choose.

Second, the calibrated weights are closely related. The warm-glow II weight, $\beta_{\mathrm{II}} = 0.896$, equals the altruist's effective weight on the log of her child's human capital, βB. It exceeds $\beta = 0.675$ because human capital is valuable to later generations as well, through the spillover ρ and through the child's own investment. The warm-glow I weight, $\beta_{\mathrm{I}} = \gamma\beta_{\mathrm{II}} = 0.111$, is an order of magnitude smaller because it applies to the input e rather than to the output h'. With a generation length of thirty years, β corresponds to an annual discount factor of about 0.987.

Third, because the three specifications share every technology parameter, the calibration absorbs none of the differences in motive. Any divergence in the counterfactuals below comes from how each specification's valuation of the child responds to policy.

4.3 Policy Experiments

We consider two permanent and unanticipated interventions, both introduced at $t = 0$ and known to apply to all future generations: an expansion of public education and a universal basic income. To compare the three specifications, it helps to write their first-order conditions in a common form. Let $\eta(h) \equiv V'(h)\, h$ denote the elasticity of the altruistic value function with respect to human capital. The envelope condition of (12) gives the recursion

$$\eta(h) \;=\; \frac{(1-\tau)h}{c(h)} \;+\; \beta\rho\, \mathbb{E}_{\varepsilon}\big[\eta(h')\big]. \tag{21}$$

For parents with $e > 0$, the first-order conditions are

$$\mathcal{N}_{\mathrm{I}}: \qquad \frac{e}{y-e} \;=\; \beta_{\mathrm{I}}, \tag{22}$$

$$\mathcal{N}_{\mathrm{II}}: \qquad \frac{\mathcal{E}+e}{y-e} \;=\; \beta_{\mathrm{II}}\,\gamma \cdot \frac{g(h,e)}{g(h,e)+\theta}, \qquad g(h,e) \equiv (1-\tau) Z\, h^{\rho} (\mathcal{E}+e)^{\gamma}, \tag{23}$$

$$\mathcal{R}: \qquad \frac{\mathcal{E}+e}{y-e} \;=\; \beta\,\gamma \cdot \mathbb{E}_{\varepsilon}\big[\eta(h')\big]. \tag{24}$$

At baseline, $\eta \equiv B$ and all three right-hand sides equal $\lambda/(1-\lambda)$. The interventions break this equality in different ways. Under warm glow I, the right-hand side is constant and the left-hand side does not involve $\mathcal{E}$. Under warm glow II, the weight on investment is scaled by the share of the child's expected income that comes from earnings. Under altruism, the weight is the expected elasticity of the child's value function, which by (21) depends on the child's consumption choice and, recursively, on the choices of all later descendants.

We report two kinds of results. The partial equilibrium (PE) analysis holds τ at its baseline value. It either evaluates the new policy functions on the baseline distribution ("on

impact") or reports the new stationary distribution. The general equilibrium (GE) analysis adjusts τ so that (11) holds in the new stationary distribution. We compute GE outcomes on grids with step 0.01 over $\mathcal{E} \in [0, 0.3]$ and $\theta \in [0, 0.3]$, and report PE and GE results at the same policy levels, $\mathcal{E} = 0.01$ and $\theta = 0.05$, which are points of these grids.[3] Appendix B reports transition paths.

Welfare measurement. We evaluate welfare with each specification's own value function, so that the planner respects the preferences that generate behavior in each model. Under altruism, the value of a parent with human capital h under policy p is her dynastic utility, $\mathcal{V}_p^{\mathcal{R}}(h) = V_p(h)$, the solution to (12) in the policy environment p. Under warm glow, it is the parent's maximized objective,

$$\mathcal{V}_p^{\mathcal{N}_j}(h) \;=\; \max_{e \in \Gamma(h)} \Big\{ \log\big(y - e\big) + \Phi_j(h, e;\, p) \Big\}, \qquad j \in \{\mathrm{I}, \mathrm{II}\}, \tag{25}$$

where the notation $\Phi_j(\cdot; p)$ makes explicit that Φ_{II} depends on the policy environment through the child's expected income. Aggregate welfare is the stationary average $W(p, m) = \int \mathcal{V}_p^m \, d\mu_p^m$, defined formally in (28) below. We express value differences as consumption equivalents: the gain ω is the proportional increase in consumption that would raise the baseline value to its value under the policy. Under altruism, the increase applies in every period to every member of the dynasty; under warm glow, it applies to the parent alone. Because values are logarithmic in consumption,

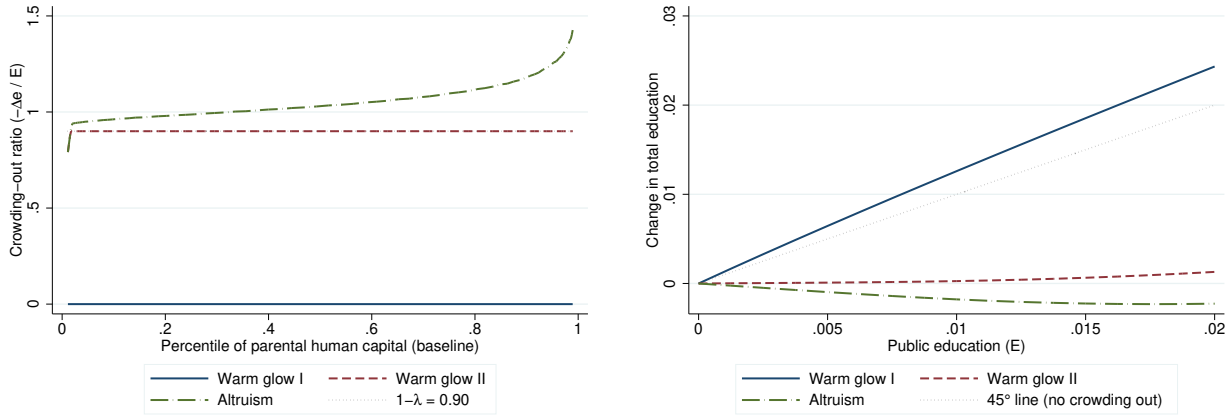
$$\omega^{\mathcal{R}} = \exp\Big((1 - \beta) \big[\mathcal{V}_p^{\mathcal{R}} - \mathcal{V}_0^{\mathcal{R}} \big] \Big) - 1, \qquad \omega^{\mathcal{N}_j} = \exp\Big(\mathcal{V}_p^{\mathcal{N}_j} - \mathcal{V}_0^{\mathcal{N}_j} \Big) - 1, \tag{26}$$

applied either to individual values or to aggregate welfare. Differences in welfare across specifications reflect both differences in behavior and differences in what parents value. For example, a policy that raises a child's expected income makes a warm-glow II parent better off even if her own consumption does not change.

4.3.1 Public Education Expansion

The first intervention introduces public education spending $\mathcal{E} > 0$ per child, financed in GE by the earnings tax. In PE, public education changes the transition function (10) but leaves the parent's budget set unchanged. It is thus a concrete example of the technology-changing interventions of Proposition 3(b). Unlike a neutral change in productivity (Example 1), an

[3] Comparisons of GE steady states abstract from the transition of the tax rate, which would require evaluating each generation's value function along the transition path.



(a) Crowding out on impact, by percentile (PE, $\mathcal{E} = 0.01$)

(b) Change in mean total education $\mathcal{E} + e$ (GE)

Figure 1: Public education under three observationally equivalent specifications

Note: Panel (a): $-\Delta e/\mathcal{E}$ at each percentile of the baseline distribution, with the post-intervention policy functions evaluated at baseline τ; the dotted line is $1 - \lambda = 0.90$. Panel (b): change in mean $\mathcal{E} + e$ across budget-balanced stationary equilibria for $\mathcal{E} \in [0, 0.3]$; the dotted line is the 45-degree line (no crowding out).

additive public input changes the marginal return to private spending, so condition (6) has no reason to hold.

Mechanisms. Under warm glow I, condition (22) does not involve $\mathcal{E}$, so private spending does not change. Crowding out is exactly zero in PE, and total education rises one-for-one with public provision, as Proposition 3(b) predicts for the warm-glow twin $\mathcal{N}(\Phi^*)$. Under warm glow II with $\theta = 0$, condition (23) has the closed-form solution

$$e^{\mathcal{N}_{\text{II}}}(h) \;=\; \max\{0,\; \lambda\, y - (1-\lambda)\,\mathcal{E}\}. \tag{27}$$

Each unit of public education therefore displaces $1 - \lambda = 0.90$ units of private spending in families that keep spending, while families with $y < (1-\lambda)\mathcal{E}/\lambda$ stop spending privately and are crowded out by less. Under altruism, condition (24) has the same form as (23), with β_{II} replaced by $\beta\,\mathbb{E}[\eta(h')]$. The difference is that the altruistic parent also anticipates that public education will let her child cut her own private spending. The child then consumes more relative to her earnings, which lowers $(1-\tau)h'/c'$ in (21). The same happens for every later generation, so the expected elasticity falls below its baseline value $B = \beta_{\text{II}}/\beta$. The altruistic parent therefore aims for a lower ratio of total education to consumption than the warm-glow II parent and cuts her private spending by more. In our calibration, this ordering holds at every percentile of the baseline distribution (Figure 5 in Appendix B).

Crowding out. Figure 1(a) shows crowding out on impact for $\mathcal{E} = 0.01$, which is about 13 percent of baseline private spending per child. Warm glow I implies no crowding out at any percentile. Warm glow II implies a ratio of 0.90 everywhere except among the poorest 2 percent of families, who stop spending privately, and the population average is also 0.90. Under altruism, the average is 1.05, so each dollar of public education reduces private spending by more than a dollar. The altruistic ratio rises with income, exceeds one above roughly the 35th percentile, and reaches about 1.4 at the top. The reason for the gradient is that a lower expected elasticity reduces the desired ratio of total education to consumption by the same proportion in every family, so the resulting cut in private spending is proportional to resources. In rich families, it exceeds the public provision. Section 4.4 shows that most of the additional crowding out under altruism arises because the parent anticipates that her own child will also be crowded out.

These predictions bear directly on empirical work on how private education spending responds to public provision (Peltzman, 1973). Holding every pre-intervention moment fixed, a measured crowding-out ratio near zero is consistent only with warm glow I, a ratio near 0.90 with warm glow II, and a ratio above one with altruism.

General equilibrium. Once the expansion is financed and the economy reaches its new steady state, the differences grow larger still (Figure 1(b) and Table 2). At $\mathcal{E} = 0.01$, warm glow I *crowds in* private spending. Mean private spending rises from 0.075 to 0.078, a stationary crowding-out ratio of -0.26. Because $e = \lambda y$ and average resources equal $\int h \, d\mu - G - \mathcal{E}$, the 3.6 percent rise in average human capital raises resources by more than the cost of the public input. The tax base expands so much that public education almost pays for itself: the budget-balancing tax rate rises by only 0.1 percentage points, compared with 0.9 and 1.1 points under warm glow II and altruism. Under those two specifications, stationary crowding out is 0.97 and 1.18, slightly above its value on impact. As a result, total education rises by 126 percent of the public input under warm glow I and by 3 percent under warm glow II, and it falls by 18 percent of the public input under altruism. Average human capital rises by 3.6 and 0.2 percent under the two warm-glow specifications and falls by 0.3 percent under altruism.

The gap between warm glow II and altruism narrows as the expansion grows. Since private spending cannot fall below zero, the crowding-out ratio is bounded by baseline private spending divided by $\mathcal{E}$, and under both specifications it falls toward this bound, reaching about one-quarter at $\mathcal{E} = 0.3$ (Figure 6). Total education and average human capital then rise steadily. At $\mathcal{E} = 0.3$, private spending has nearly vanished under both specifications, and their tax rates and average human capital agree to within 0.1 percent. Once private spending

reaches zero, the motive no longer affects behavior. The disagreement between warm glow II and altruism is therefore concentrated in small and moderate expansions, which is also the range that empirical estimates of crowding out typically cover. Warm glow I stays apart throughout: it crowds in private spending at every $\mathcal{E}$, and human capital rises by more.

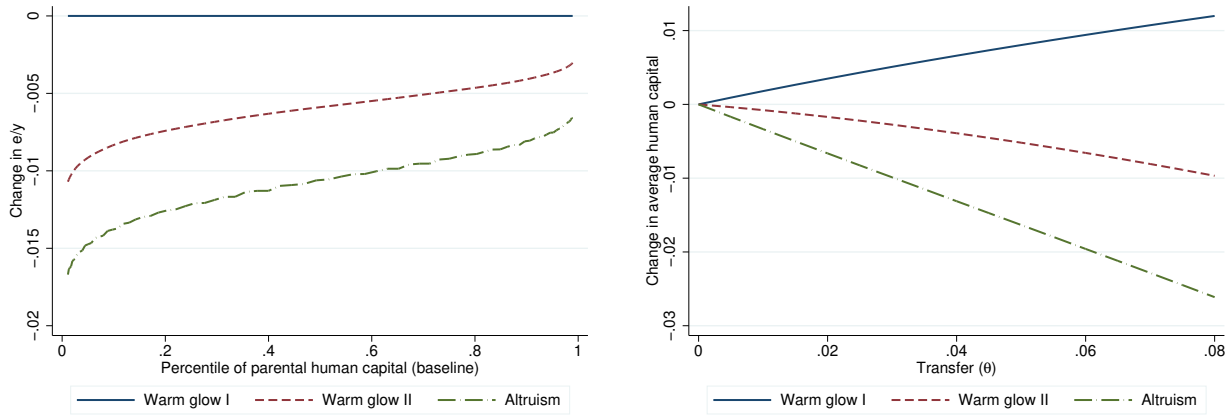
Welfare. At $\mathcal{E} = 0.01$, the GE welfare gains are 4.5, 0.8, and 1.3 percent in consumption equivalents under warm glow I, warm glow II, and altruism, and they arise for different reasons. Under warm glow I, the entire gain comes from higher human capital. The generation alive at the reform gains nothing in PE, because neither its consumption nor its warm glow changes, and the gains accrue to later generations as human capital builds up (Appendix B). Under altruism, average human capital and total education barely move for small expansions, so the gain is redistributive. Public provision is the same for every child and is financed by a proportional tax, and crowding out converts it into consumption, so it works much like a lump-sum transfer. Under warm glow II, the parent values her child's expected income, which hardly changes, while her taxes rise.

Across the full range, welfare is hump-shaped under all three specifications (Figure 6). The welfare-maximizing levels are 0.130 under warm glow I, 0.178 under warm glow II, and 0.184 under altruism (Table 4). Warm glow I calls for the least public education. Because private spending is crowded in, total education is already high at modest levels of $\mathcal{E}$, and diminishing returns set in sooner. Under the other two specifications, public spending mostly replaces private spending, so a larger public input is needed before total education rises. The specifications disagree sharply about behavior, yet their optimal policies are fairly close, a point we return to in Section 5.

4.3.2 Universal Basic Income

The second intervention pays a lump-sum transfer $\theta > 0$ to every parent. It raises resources (9) by the same amount for everyone, so in proportional terms it matters most at the bottom of the income distribution. Because the policy is permanent, the parent also knows that her child, and every later descendant, will receive θ.

Mechanisms. The three specifications respond differently to this last fact. Under warm glow I, condition (22) is unaffected. The parent spends the same fraction λ of her now higher resources on education, so private investment rises by $\lambda\theta$ and the education share is unchanged. Under warm glow II, the transfer the child will receive reduces the share of the child's expected income that depends on her human capital. The factor $g/(g+\theta)$ in (23) falls, and with it the education share, most of all for low-income parents, whose children's expected



(a) Change in e/y by percentile (PE, $\theta = 0.05$)

(b) Change in mean human capital (GE)

Figure 2: Universal basic income under three observationally equivalent specifications

Note: Panel (a): change in the education share at each percentile of the baseline distribution, at baseline τ. Panel (b): change in mean human capital across budget-balanced stationary equilibria for $\theta \in [0, 0.3]$.

earnings g are smallest relative to θ. Under altruism, the transfer raises the consumption of the child and every later descendant relative to their earnings. This lowers $(1-\tau)h/c$ in (21) for every future generation, and through it the elasticity η and the right-hand side of (24). The altruistic parent thus takes into account the transfers received by all later generations and their investment responses, whereas the warm-glow II parent considers only her child's income.

Results. Figure 2(a) shows how the education share responds to a transfer of $\theta = 0.05$, about seven percent of mean after-tax earnings. Under warm glow II, the share falls by 1.1 percentage points at the bottom and by 0.3 points at the top. Under altruism, it falls by more at every percentile, from 1.7 points at the bottom to 0.7 at the top. In the PE stationary distribution, the mean education share falls by 0.61 points under warm glow II and by 1.08 points under altruism. Taking into account the transfers of all future generations thus nearly doubles the investment response.

Once the transfer is financed, the specifications disagree even on the sign of its effect on human capital, at every transfer level we consider (Figure 2(b)). At $\theta = 0.05$, average human capital rises by 0.8 percent under warm glow I and falls by 0.5 and 1.6 percent under warm glow II and altruism. At $\theta = 0.3$, the changes are $+3.1$, -7.8, and -12.4 percent. The increase under warm glow I reflects diminishing returns. Under budget balance, average resources are unchanged at $\int h\, d\mu - G$ to first order, but the transfer shifts them toward low-income families. At baseline, the marginal product of education spending, $\gamma\, \mathbb{E}[h']/e$, is

proportional to $h^{\rho+\gamma-1}$, so it is highest in exactly these families. Under warm glow II and altruism, the fall in the education share dominates. The resulting erosion of the tax base requires larger tax increases: at $\theta = 0.05$, the budget-balancing tax rate rises to 0.298, 0.302, and 0.305, and welfare rises by 8.2, 6.5, and 5.4 percent in consumption equivalents.

Optimal transfer. Under warm glow II and altruism, welfare is hump-shaped in θ, with optima at 0.220 and 0.264 (Figure 7 and Table 4). Human capital falls most under altruism, yet the altruistic optimum is the larger of the two. The altruistic parent values the insurance that the transfer provides against her descendants' shocks, while the warm-glow II parent considers only her child's expected income. Under warm glow I, welfare rises throughout the range, because the budget-balanced transfer has no efficiency cost under this specification. With $e = \lambda y$, the parent's value is $(1+\beta_{\mathrm{I}}) \log y$ plus a constant, so aggregate welfare depends only on $\int \log y \, d\mu$. The transfer raises this integral both by equalizing resources and by raising average human capital, and neither effect reverses within the range we consider.

Unlike public education, the UBI changes the flow payoff and the feasible set rather than the technology. Warm glow I and warm glow II coincide everywhere in the baseline environment, yet they disagree about the sign of the UBI's effect on human capital. Their disagreement does not come from differences away from observed choices, the channel isolated in Lemma 1(b), which holds the warm-glow function fixed. It comes instead from how the two payoffs depend on the policy environment, which pre-intervention data reveal even less. For a budget-changing policy, the choice of warm-glow specification alone can thus reverse the sign of a counterfactual.

4.4 Foresight Content of the Interventions

Where does the disagreement between warm glow II and altruism come from? To find out, we compute the depth hierarchy (7) for each intervention in PE, with terminal value $\phi(h) = B \log\big((1-\tau)h+\theta\big)$.[4] In the baseline environment, ϕ equals the altruistic value function up to a constant, so every depth-k model generates the common baseline policy. After an intervention, the depth-zero parent values her child's resources with the pre-intervention elasticity B. For public education ($\theta = 0$), $\mathbb{E}\log((1-\tau)h')$ and $\log((1-\tau)\mathbb{E}[h'])$ differ only by a constant, and $\beta B = \beta_{\mathrm{II}}$, so the depth-zero model coincides exactly with warm glow II. For the UBI, the two differ only in the order of the expectation and the logarithm. Each additional level of depth lets one more generation re-optimize in response to the policy.

Convergence is fast (Figure 3). A single generation of foresight closes 77 percent of the

[4]This terminal value is unbounded, like log utility itself; on the compact grid used in the computations the conditions of Proposition 4 hold.

Table 2: Counterfactual Predictions of Three Observationally Equivalent Specifications

	Warm glow I	Warm glow II	Altruism
A. Public education, $\mathcal{E} = 0.01$			
Crowding out on impact (PE)	0.00	0.90	1.05
Crowding out, stationary (GE)	−0.26	0.97	1.18
Change in total education, % of $\mathcal{E}$ (GE)	126	3	−18
Budget-balancing tax rate (GE)	0.251	0.259	0.261
Change in mean h, % (GE)	3.6	0.2	−0.3
Welfare gain, % cons. equiv. (GE)	4.5	0.8	1.3
B. Universal basic income, $\theta = 0.05$			
Change in mean e/y, pp (PE, stationary)	0.00	−0.61	−1.08
Budget-balancing tax rate (GE)	0.298	0.302	0.305
Change in mean h, % (GE)	0.8	−0.5	−1.6
Welfare gain, % cons. equiv. (GE)	8.2	6.5	5.4

Note: PE holds $\tau = 0.25$; GE adjusts τ to balance (11) in the new stationary distribution. PE and GE results refer to the same policy levels, which are points of the GE grids. Crowding out is $-\Delta e/\mathcal{E}$, averaged over the baseline distribution (on impact) or computed from stationary means (GE). Welfare gains apply (26) to aggregate welfare $W(p, m)$; under altruism the gain is a permanent increase in every generation's consumption. Welfare-maximizing policies are reported in Table 4.

policy gap for the UBI and 61 percent for public education, and the foresight content is $\mathcal{F}(0.1) = 2$ for the UBI and 3 for public education.[5] By contrast, the geometric factor β^k in the value-function bound of Proposition 4 falls below 10 percent only after six generations. Policies converge faster than this bound because choices depend on the marginal value of the child's human capital, not its level. By (21), the response of each generation beyond the child enters the parent's marginal valuation with an additional factor of $\beta\rho \approx 0.16$, rather than β.

The disagreement between warm glow II and altruism is therefore about whether the parent anticipates that her child and grandchild will re-optimize, not about the infinite horizon. In the public education experiment, a depth-one parent already anticipates that her own child will be crowded out, and this alone accounts for most of the additional crowding out under altruism.

5 Minimax-Regret Policy Evaluation

The previous sections show that pre-intervention data cannot settle which motive is right, and that the answer matters for predicted behavior. A planner must nonetheless choose a

[5] With a baseline-weighted L^1 norm instead of the supremum norm, $\mathcal{F}(0.1) = 2$ for both interventions.

Table 3: Behavioral Responses Implied by the Three Specifications

	Warm glow I	Warm glow II	Altruism
Child valued through	Own investment e	Child's expected income	Child's value $V(h')$
Position in depth hierarchy	Outside (Φ indep. of T)	Depth 0 (exact if $\theta = 0$)	Depth ∞
Welfare evaluated with	$\log c + \beta_{\mathrm{I}} \log e$	$\log c + \Phi_{\mathrm{II}}$	$V(h)$
UBI: education share	Unchanged	Falls, most at bottom	Falls more, at every percentile
Public education: crowding out on impact	0	0.90 (unconstrained)	1.05 on average; above 1 beyond the 35th percentile

Note: Partial-equilibrium responses, starting from the common baseline of Lemma 2. Magnitudes for altruism, and for warm glow II under the UBI, are computed numerically. Warm glow II coincides exactly with the depth-zero model of Section 4.4 for public education and approximately for the UBI.

policy. This section asks how she should do so when she cannot rule out any of the admissible models. We set up a minimax-regret criterion, show that recursion creates a systematic asymmetry in regret, and apply the criterion to the two interventions of Section 4.3.

5.1 Setup

A planner chooses a policy p from a compact interval $\mathcal{P} \subset \mathbb{R}$ without knowing the true model of parental behavior. The set of admissible models is $\mathcal{M} = \{\mathcal{R}\} \cup \{\mathcal{N}(\Phi) : \Phi \in \mathcal{C}\}$, where $\mathcal{R}$ is the altruistic model with the discount factor β implied by the Euler equation (3), and $\mathcal{C}$ is the set of warm-glow functions consistent with the pre-intervention data.

For each model $m \in \mathcal{M}$ and policy $p \in \mathcal{P}$, the equilibrium determines a policy function σ_p^m, a stationary distribution μ_p^m over human capital, and a value function $\mathcal{V}_p^m$ that gives the maximized objective of a parent with human capital h. Under $\mathcal{R}$, $\mathcal{V}_p^{\mathcal{R}} = V_p$ is the dynastic value function solving (12) under policy p. Under $\mathcal{N}(\Phi)$, $\mathcal{V}_p^{\mathcal{N}}(h) = u\big(c(h; \sigma_p^{\mathcal{N}}, p)\big) + \Phi\big(h, \sigma_p^{\mathcal{N}}(h); p\big)$ is the parent's maximized static objective, as in (25). Social welfare is the average value in the stationary distribution,

$$W(p, m) \;=\; \int \mathcal{V}_p^m(h)\, d\mu_p^m(h). \tag{28}$$

This criterion is welfarist: the planner evaluates outcomes with the same preferences that generate behavior in each model. Under altruism, it is the average dynastic utility of the stationary population, and under warm glow, the average of parents' own objectives, including

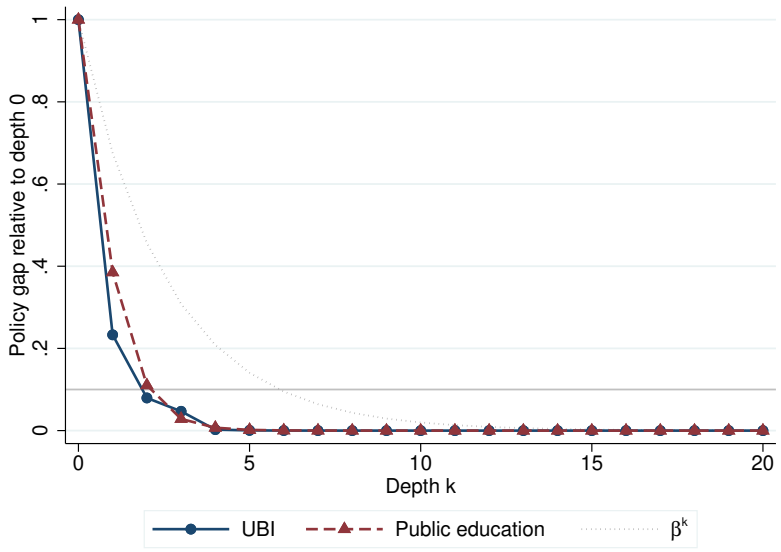


Figure 3: Foresight content: policy gap relative to depth zero

Note: $\|\tilde{\sigma}_k - \tilde{\sigma}^{\mathcal{R}}\|_\infty / \|\tilde{\sigma}_0 - \tilde{\sigma}^{\mathcal{R}}\|_\infty$, with the supremum taken over the support of the baseline distribution, for $\theta = 0.05$ (UBI) and $\mathcal{E} = 0.01$ (public education) at baseline τ. The solid horizontal line marks $\epsilon = 0.1$; the dotted curve is β^k.

the warm-glow payoff.

The optimal policy under model m is $p^*(m) = \arg\max_{p \in \mathcal{P}} W(p, m)$. The *regret* from choosing p when the true model is m, the *maximum regret* of p, and the *minimax-regret policy* are

$$\begin{aligned} \mathrm{Reg}(p, m) &= W\big(p^*(m), m\big) - W(p, m), \qquad \mathrm{MaxReg}(p) = \sup_{m \in \mathcal{M}} \mathrm{Reg}(p, m), \\ p^{MR} &= \arg\min_{p \in \mathcal{P}} \mathrm{MaxReg}(p). \end{aligned} \tag{29}$$

Minimax regret does not require the planner to assign probabilities to models. This is an advantage here, since $\mathcal{M}$ contains structurally different specifications over which probability judgments would be hard to defend.

5.2 Theoretical Properties

In utility units, recursion makes welfare more sensitive to policy mistakes. Under $\mathcal{R}$, each parent's value includes the discounted values of all her descendants, so a policy that distorts the stationary allocation is penalized once for every generation that inherits the distortion. Under $\mathcal{N}$, a parent's value includes only her own consumption and warm-glow payoff. A policy mistake shifts the stationary distribution in both models, but only under recursion does each parent's welfare add up the losses of her entire dynasty. The next result makes

this precise and shows how it shapes the minimax-regret policy. We call $W(\cdot, m)$ *strictly single-peaked* if it is strictly increasing below its maximizer and strictly decreasing above it.

Proposition 5 (Regret amplification under recursion)**.** *Let $\mathcal{M} = \{\mathcal{R}, \mathcal{N}(\Phi)\}$. For each model m and policy p, suppose μ_p^m is a stationary distribution with $\int |\mathcal{V}_p^m| \, d\mu_p^m < \infty$, and let $W_{\text{flow}}(p, m) = \int u\big(c(h; \sigma_p^m, p)\big) \, d\mu_p^m(h)$ denote average flow utility.*

(a) Under $\mathcal{R}$, welfare is proportional to average flow utility: $W(p, \mathcal{R}) = W_{\text{flow}}(p, \mathcal{R})/(1-\beta)$.

(b) Under $\mathcal{N}(\Phi)$, $W(p, \mathcal{N}) = W_{\text{flow}}(p, \mathcal{N}) + \bar{\Phi}(p)$, with $\bar{\Phi}(p) = \int \Phi\big(h, \sigma_p^{\mathcal{N}}(h); p\big) \, d\mu_p^{\mathcal{N}}(h)$.

(c) Suppose $W(\cdot, \mathcal{R})$ and $W(\cdot, \mathcal{N})$ are continuous and strictly single-peaked on $\mathcal{P}$, with distinct maximizers. Then the minimax-regret policy is unique, lies strictly between $p^*(\mathcal{N})$ and $p^*(\mathcal{R})$, and equalizes the two regrets.

(d) Suppose, in addition, that the maximizers are interior, that $W(\cdot, m)$ is twice differentiable, and that $a_m \equiv |W''(p^*(m), m)| > 0$. If each regret is replaced by its second-order approximation $\frac{1}{2} a_m (p - p^*(m))^2$, then

$$\frac{\text{Reg}\big(p^*(\mathcal{N}), \mathcal{R}\big)}{\text{Reg}\big(p^*(\mathcal{R}), \mathcal{N}\big)} = \frac{a_{\mathcal{R}}}{a_{\mathcal{N}}}, \qquad \frac{|p^{MR} - p^*(\mathcal{R})|}{|p^{MR} - p^*(\mathcal{N})|} = \sqrt{\frac{a_{\mathcal{N}}}{a_{\mathcal{R}}}},$$

and, by part (a), $a_{\mathcal{R}} = |\partial^2 W_{\text{flow}}(p^*(\mathcal{R}), \mathcal{R})/\partial p^2|/(1-\beta)$.

Part (a) rests on a simple observation. In a stationary distribution, the descendants of the population are distributed exactly like the population itself, so average dynastic value equals average flow utility times $\sum_{t \geq 0} \beta^t = 1/(1-\beta)$. Any distortion of stationary flow utility is therefore counted once for every generation it affects. Under $\mathcal{N}$, no continuation term adds up distortions across generations, which gives part (b). Part (c) says that, with two models, the robust policy is a compromise between their optima. Part (d) says where the compromise lies: the robust policy is pulled toward the model whose welfare is more sharply curved at its optimum. By part (a), the curvature of altruistic welfare scales with $1/(1-\beta)$. Holding the curvature of flow utility and of the warm-glow payoff fixed, the pull is therefore toward altruism once β is close enough to one. This is a sufficient condition rather than a necessary one, and in our application $a_{\mathcal{R}} > a_{\mathcal{N}}$ already holds at the calibrated β (Section 5.3).

Remark 1 (Cardinal comparability)**.** Because minimax regret compares welfare losses across models, its answer depends on how welfare is scaled in each model. Under (28), altruistic welfare adds up the utility of a parent's entire dynasty, whereas warm-glow welfare covers

only her own objective. Part of the amplification in Proposition 5(a) therefore reflects a difference in what each model's welfare measures. An alternative is to express regret in consumption equivalents using (26), which removes the factor $1/(1-\beta)$ from altruistic regret. Any remaining asymmetry must then come from the curvature of stationary flow utility, from the warm-glow term, and from differences in how the stationary distribution responds to policy across models.

The depth hierarchy offers a complementary bound. A depth-k parent's maximized objective is $\tilde{V}_{k+1}$, so for any fixed distribution μ, $|\int \tilde{V}_{k+1}\,d\mu - \int \tilde{V}\,d\mu| \le \|\tilde{V}_{k+1} - \tilde{V}\|_\infty$, and Proposition 4(c) bounds the welfare difference between depth-k and altruistic predictions at the same geometric rate when both are evaluated on the same population, such as the generation alive at the reform. Comparisons of stationary welfare also reflect differences between the models' stationary distributions, which this bound does not cover. The bound also says nothing about warm-glow specifications outside the hierarchy, such as warm glow I.

5.3 Application to Intergenerational Policies

We now apply the framework to the two interventions, with $\mathcal{M} = \{\mathcal{R}, \mathcal{N}_{\mathrm{I}}, \mathcal{N}_{\mathrm{II}}\}$ and budget-balanced stationary equilibria for $\mathcal{E} \in [0, 0.3]$ and $\theta \in [0, 0.3]$. By Lemma 1(b), the full set of warm-glow functions consistent with the data is larger, so restricting attention to three exactly equivalent specifications yields a lower bound on maximum regret. Following Remark 1, we report regret both in utility units, as in (29), and in consumption equivalents,

$$\mathrm{Reg}^{\mathrm{CE}}(p, m) \;=\; \exp\big(\kappa_m \,\mathrm{Reg}(p, m)\big) - 1, \qquad \kappa_{\mathcal{R}} = 1 - \beta, \quad \kappa_{\mathcal{N}_j} = 1. \tag{30}$$

By (26), this is the permanent proportional consumption loss from choosing p rather than $p^*(m)$. To second order, the curvature of consumption-equivalent regret at the optimum is $\kappa_m a_m$. Table 4 summarizes the results, and Figure 4 plots the regret curves.

Public education. All three welfare functions have interior optima, ordered $p^*(\mathcal{N}_{\mathrm{I}}) < p^*(\mathcal{N}_{\mathrm{II}}) < p^*(\mathcal{R})$. In utility units, altruistic welfare is the most sharply curved of the three, so $a_{\mathcal{R}} > a_{\mathcal{N}}$ holds at the calibrated $\beta = 0.675$, well below one. Regret binds for the two extreme models, warm glow I and altruism, and the minimax-regret policy, $\mathcal{E} \approx 0.164$, lies about 60 percent of the way from the warm-glow-I optimum to the altruistic one, as the second-order formula of Proposition 5(d) predicts.

The picture changes in consumption equivalents. Once the factor $1/(1-\beta) \approx 3.1$ is removed, altruistic welfare becomes the *least* curved of the three: average flow utility under

Table 4: Optimal and Minimax-Regret Policies

	Warm glow I	Warm glow II	Altruism
A. Public education, $\mathcal{E}$			
Welfare-maximizing policy $p^*(m)$	0.130	0.178	0.184
Curvature a_m, utility units	6.6	9.9	15.6
Curvature $\kappa_m a_m$, consumption equivalents	6.6	9.9	5.1
Regret of no intervention, % cons. equiv.	16.2	21.2	16.0
Minimax-regret policy, utility units	0.164	(binding: warm glow I, altruism)	
Minimax-regret policy, cons. equivalents	0.157	(binding: warm glow I, warm glow II)	
Maximum regret at this policy, % cons. equiv.		0.2	
B. Universal basic income, θ			
Welfare-maximizing policy $p^*(m)$	≥ 0.300	0.220	0.264
Curvature a_m, utility units	–	6.3	18.3
Curvature $\kappa_m a_m$, consumption equivalents	–	6.3	5.9
Regret of no intervention, % cons. equiv.	≥ 35.3	15.7	15.9
Minimax-regret policy, all three models	0.280	(binding: warm glow I, warm glow II)	
Maximum regret at this policy, % cons. equiv.		1.3	
Minimax-regret policy, warm glow II and altruism	0.246 (utility units),	0.240 (cons. equivalents)	

Note: Budget-balanced stationary equilibria. Each optimum $p^*(m)$ is found by golden-section search between the neighbors of the best point of a grid with step 0.01, solving the GE model at every trial policy; $a_m = |W''(p^*(m), m)|$ is a central second difference with step 0.01. Minimax-regret policies minimize (29) or (30) by golden-section search, solving all models at every trial policy (Appendix C). Under warm glow I, UBI welfare rises up to the upper bound of the range, so its optimum and curvature are not defined within it. With all three models, the binding UBI specifications share the same welfare scale, so the minimax-regret transfer is the same in both units.

altruism is less sensitive to public education than either warm-glow objective. The amplification in utility units is therefore entirely a consequence of the dynastic normalization discussed in Remark 1. The minimax-regret policy falls to $\mathcal{E} \approx 0.157$, roughly midway between the warm-glow-I and altruistic optima. The binding models are now the two warm-glow specifications, with altruistic regret close behind, and all three regret curves nearly intersect at this point (Figure 4(b)).

Universal basic income. Under warm glow I, the transfer has no efficiency cost (Section 4.3.2), so its optimum lies at the upper bound of the range and part (d) of Proposition 5 does not apply. With all three models, regret binds for the two warm-glow specifications. Because these share the same welfare scale, the minimax-regret transfer, $\theta \approx 0.28$, is the same in both units. Altruism never binds, even though its optimum lies between the other two. The robust transfer depends on the upper bound of the range: extending the range

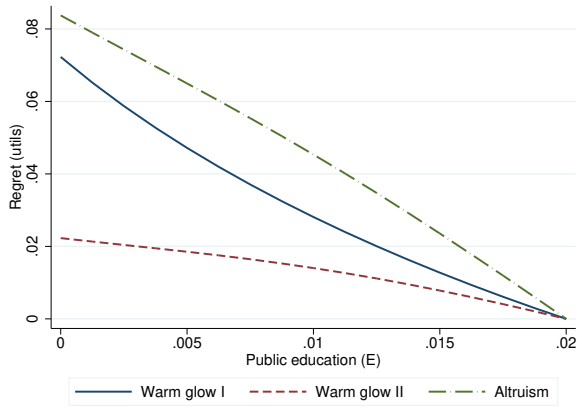
would raise the warm-glow-I optimum and its regret at every θ, and so push the robust transfer up. If we restrict attention to the two specifications with interior optima, warm glow II and altruism, the pattern of public education reappears. In utility units, altruistic welfare is much more curved, and the minimax-regret transfer, $\theta \approx 0.246$, lies about 60 percent of the way toward the altruistic optimum. In consumption equivalents, altruistic welfare is slightly flatter, and the minimax-regret transfer, $\theta \approx 0.240$, lies close to the midpoint.

How much is at stake. Uncertainty about the motive matters much more for predicted behavior than for the welfare cost of a robust policy. The regret of not intervening at all is between 16 and 21 percent of consumption for public education, and between 16 and at least 35 percent for the UBI. The maximum regret of the minimax-regret policy, in consumption equivalents, is only about 0.2 percent for public education and about 1.3 percent for the UBI. For public education, even adopting one model's optimum when another model is true costs less than 1.5 percent of consumption, although the models disagree about whether public provision crowds private spending in, out, or out by more than one-for-one. The UBI under warm glow I is the exception. Because this specification removes the efficiency cost of redistribution, adopting another model's optimum costs several percent of consumption. The welfare-maximizing policy is thus sensitive to the motive mainly when the motive decides whether the policy has an efficiency cost at all.

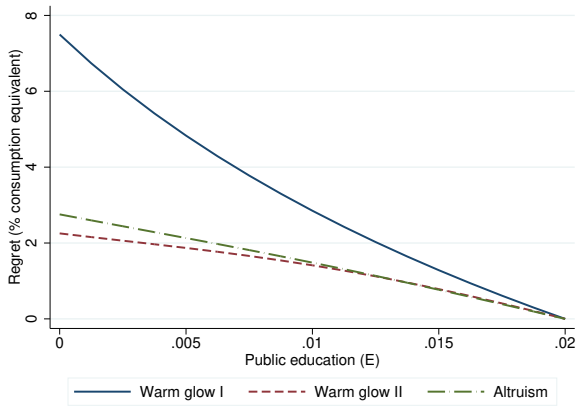
6 Conclusion

This paper has argued that the choice between altruism and warm glow matters for policy evaluation in ways that pre-intervention data cannot settle. Any behavior generated by altruistic parents can be reproduced by warm-glow parents, yet the two models agree about a counterfactual only if the intervention happens to leave the altruist's marginal value of investing unchanged. The altruistic model is testable and its counterfactuals are determinate. The warm-glow model can fit any behavior, but only because it leaves counterfactuals open. The foresight hierarchy connects the two and shows how the disagreement builds up as parents anticipate more generations of re-optimization.

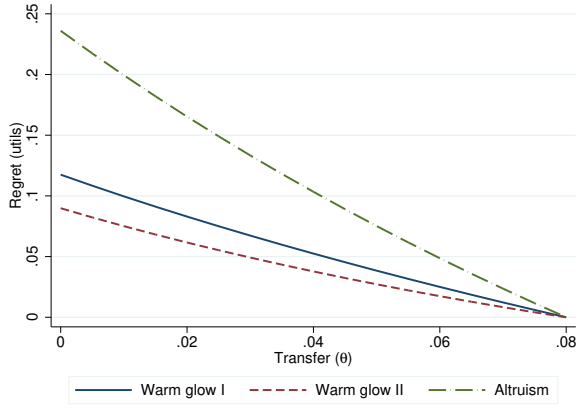
Our quantitative application shows that the problem is not merely theoretical. Altruism and two warm-glow specifications, calibrated to be exactly observationally equivalent, disagree about whether public education crowds out private spending and by how much, and about whether a budget-balanced universal basic income raises or lowers average human capital. The gap between the income-based warm-glow specification and altruism closes within a few generations of foresight, so the key modeling question is whether parents anticipate



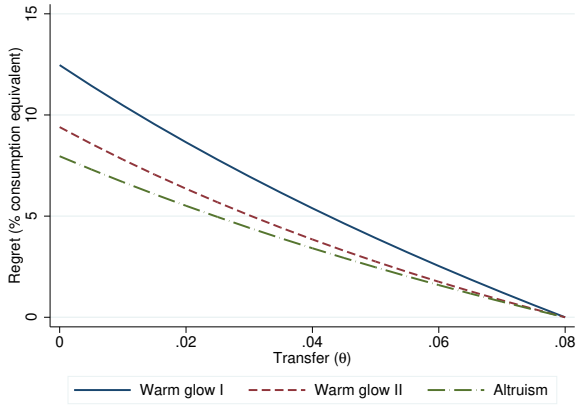
(a) Public education, utils



(b) Public education, % consumption equivalent



(c) UBI, utils



(d) UBI, % consumption equivalent

Figure 4: Regret $\mathrm{Reg}(p, m)$ across policies

Note: Regret (29) and (30), relative to each model's welfare-maximizing policy, computed from budget-balanced stationary equilibria on a grid with step 0.01 over $[0, 0.3]$. Under warm glow I, UBI welfare is maximized at the upper bound of the range.

how their children will respond, not whether they look infinitely far ahead. When the planner chooses policy by minimax regret, recursion pulls the robust policy toward the altruistic optimum in utility units, as the theory predicts. This pull reflects how dynastic welfare is normalized, however, and it disappears in consumption equivalents. Overall, the specifications disagree far more about behavior than about the welfare cost of a robust policy, except where one of them removes the efficiency cost of the policy altogether.

Our model is deliberately simple, and relaxing it would sharpen these conclusions. With non-logarithmic preferences or a labor supply margin, redistribution would carry efficiency costs under every specification; their absence under warm glow I is what places its optimal transfer at the edge of the policy range. Richer technologies of skill formation (Cunha

and Heckman, 2007), with imperfect substitution between public and private inputs, would change how crowding out feeds into welfare. Computing the robust policy over the full set of warm-glow functions consistent with the data, rather than over three specifications, would give a tighter answer to what a cautious planner should do. On the empirical side, quasi-experimental variation in public education spending can help discriminate between motives. A crowding-out ratio near zero is hard to reconcile with either the income-based warm-glow specification or altruism, while telling those two apart requires precise estimates from modest expansions, before private spending reaches zero and the two specifications coincide. Finally, the same logic applies to other channels through which parents shape their children's prospects, such as fertility, neighborhood choice, and peer effects.

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A Proofs

Throughout, we use two elementary facts about a function f that is concave and differentiable on a convex set S. First, if f attains its maximum over S at an interior point a, then $\nabla f(a) = 0$. Second, if $\nabla f(a) = 0$ at some $a \in S$, then $f(b) \leq f(a) + \nabla f(a) \cdot (b - a) = f(a)$ for all $b \in S$, so a maximizes f over S, uniquely if f is strictly concave.

Proof of Proposition 1. Part (a). By Assumption 1(i), V is bounded and continuous, and $T(\cdot, \cdot, \varepsilon)$ is continuous for each ε, so $\Phi^*(x, a) = \beta\, \mathbb{E}[V(T(x, a, \varepsilon))]$ is continuous by bounded convergence. By part (iii), Φ^* is continuously differentiable in a and $u(x, \cdot) + \Phi^*(x, \cdot)$ is strictly concave on $\Gamma(x)$, so Φ^* is admissible. The objective of $\mathcal{N}(\Phi^*)$ is the right-hand side of (1), so the two models have the same unique maximizer at every state: $\sigma^{\mathcal{N}} = \sigma^{\mathcal{R}} \equiv \sigma$. Both models then generate the Markov process $x_{t+1} = T(x_t, \sigma(x_t), \varepsilon_t)$ with $a_t = \sigma(x_t)$ and the same shock distribution, so they generate the same joint distribution of $(x_t, a_t)_{t \geq 0}$ for every x_0.

Part (b). Let $(x, \sigma(x))$ be interior. Since $\sigma(x)$ is an interior maximizer of the differentiable function $u(x, \cdot) + \Phi^*(x, \cdot)$, its gradient vanishes there. If $\nabla_a \hat{\Phi}(x, \sigma(x)) = \nabla_a \Phi^*(x, \sigma(x))$, the gradient of $u(x, \cdot) + \hat{\Phi}(x, \cdot)$ also vanishes at $\sigma(x)$. Since this function is strictly concave, $\sigma(x)$ is its unique maximizer over $\Gamma(x)$.

For $s > 0$, $\Phi_s = (1 + s)\Phi^* + s\, u$ is continuous and continuously differentiable in a, and $u + \Phi_s = (1 + s)(u + \Phi^*)$ is strictly concave in a with the same maximizer as $u + \Phi^*$ at every state, interior or not. So Φ_s is admissible and generates σ. For $a \in \Gamma(x)$ with $a \neq \sigma(x)$, $\nabla_a \Phi_s(x, a) - \nabla_a \Phi^*(x, a) = s\, \nabla_a (u + \Phi^*)(x, a)$. If this were zero, the second fact above would make a a maximizer of $u(x, \cdot) + \Phi^*(x, \cdot)$, contradicting uniqueness. Hence the two gradients differ at every feasible $a \neq \sigma(x)$. □

Proof of Proposition 2. Part (a). Write $\sigma = \sigma^{\mathcal{R}}$. *First-order condition.* Since (x_t, a_t) is interior and a_t maximizes the differentiable function $a \mapsto u(x_t, a) + \beta\, \mathbb{E}[V(T(x_t, a, \varepsilon))]$, and derivatives may be taken inside the expectation,

$$u_a(x_t, a_t) + \beta\, \mathbb{E}\big[V'\big(T(x_t, a_t, \varepsilon)\big)\, T_a(x_t, a_t, \varepsilon)\big] = 0. \tag{31}$$

Envelope condition. Let (x', a') be interior with $a' = \sigma(x')$. Since a' lies in the interior of the interval $\Gamma(x')$, there are $b_0 < a' < b_1$ in $\Gamma(x')$. By lower hemicontinuity of Γ, for z near x' there are points of $\Gamma(z)$ near b_0 and near b_1, and by convexity $a' \in \Gamma(z)$. For such z, define $G(z) = u(z, a') + \beta\, \mathbb{E}[V(T(z, a', \varepsilon))]$. Then $G(z) \leq V(z)$, with equality at $z = x'$. Both functions are differentiable at the interior point x', and $V - G$ has a local minimum

there, so $V'(x') = G'(x')$:

$$V'(x') = u_x(x', a') + \beta\,\mathbb{E}\big[V'\big(T(x', a', \varepsilon')\big)\,T_x(x', a', \varepsilon')\big].$$

Since $T_x = \chi\,T_a$ with χ independent of the shock, the expectation equals $\chi(x', a')\,\mathbb{E}[V'(T(x', a', \varepsilon'))\,T_a(x', a', \varepsilon$ which by (31) at (x', a') equals $-\chi(x', a')\,u_a(x', a')/\beta$. Hence $V'(x') = u_x(x', a') - u_a(x', a')\chi(x', a') = R(x', a')$.

Combining. For almost every realization of ε_t, (x_{t+1}, a_{t+1}) is interior, so $V'(x_{t+1}) = R(x_{t+1}, a_{t+1})$. Substituting into (31) gives (3).

Part (b). Let σ be continuous with $\sigma(x) \in \Gamma(x)$, and define $\Phi^\sigma(x, a) = -u(x, a) - \|a - \sigma(x)\|^2$. This function is continuous and continuously differentiable in a, and $u(x, \cdot) + \Phi^\sigma(x, \cdot) = -\|\cdot - \sigma(x)\|^2$ is strictly concave with unique maximizer $\sigma(x)$ on $\Gamma(x)$. Hence Φ^σ is admissible and generates σ. □

Proof of the claims in Example 1. Let $\mu_\varepsilon = \mathbb{E}[\log \varepsilon]$.

Claim (i). Fix $\beta \in (0, 1)$ and guess $V(h) = A + B \log h$. For $a \in (0, wh)$, the right-hand side of the Bellman equation is

$$\log(wh - a) + \beta\big[A + B(\log Z + \rho \log h + \gamma \log a + \mu_\varepsilon)\big].$$

For $B > 0$, this is strictly concave in a and tends to $-\infty$ at both ends of $(0, wh)$, so it has a unique maximizer, given by $a/(wh - a) = \beta B\gamma$, that is, $a = \lambda wh$ with $\lambda = \beta B\gamma/(1 + \beta B\gamma)$. Substituting, the coefficient on $\log h$ is $1 + \beta B(\rho + \gamma)$, so the guess is verified with $B = 1/(1 - \beta(\rho + \gamma)) > 0$ and a constant A determined by the remaining terms. With this B, $\lambda = \beta\gamma/(1 - \beta(\rho + \gamma) + \beta\gamma) = \beta\gamma/(1 - \beta\rho)$.

Because utility is unbounded, we verify that V is the value function. Under the constant-share policy, $\log h_{t+1} = k + (\rho + \gamma) \log h_t + \log \varepsilon_t$ for a constant k, so $\mathbb{E}[\log h_t]$ converges and $\beta^T \mathbb{E}[V(h_T)] \to 0$. Iterating the Bellman equation, which holds with equality along this policy, gives expected lifetime utility $\lim_T \big(V(h_0) - \beta^T \mathbb{E}[V(h_T)]\big) = V(h_0)$.

Now take any feasible plan. Iterating the Bellman inequality $u(h_t, a_t) + \beta\,\mathbb{E}_t[V(h_{t+1})] \le V(h_t)$ gives $\mathbb{E} \sum_{t<T} \beta^t u_t \le V(h_0) - \beta^T \mathbb{E}[V(h_T)]$. To bound the tail, note that $a_t \le wh_t$ implies $\log h_{t+1} \le k_0 + (\rho + \gamma) \log h_t + \log \varepsilon_t$ with $k_0 = \log Z + \gamma \log w$, so $\mathbb{E}[\log h_{T+s}] \le K + (\rho + \gamma)^s\,\mathbb{E}[\log h_T]$ with $K = |k_0 + \mu_\varepsilon|/(1 - \rho - \gamma)$. Since $u_t \le \log(wh_t)$,

$$\sum_{s \ge 0} \beta^{T+s}\,\mathbb{E}[u_{T+s}] \;\le\; \beta^T \left[\tfrac{\log w + K}{1 - \beta} + B\,\mathbb{E}[\log h_T]\right] \;=\; \beta^T \big[C + \mathbb{E}[V(h_T)]\big],$$

with $C = (\log w + K)/(1 - \beta) - A$. The same bound shows that the positive parts of utility

are summable, so expected lifetime utility is well defined. If $\mathbb{E}[\log h_T] = -\infty$, the tail, and hence the plan's value, is $-\infty$. Otherwise, adding the two bounds gives expected lifetime utility at most $V(h_0)+\beta^T C$ for every T, hence at most $V(h_0)$. So V is the value function and the constant-share policy is optimal. It is the unique optimal policy: a plan that deviates from it with positive probability at some date t makes the Bellman inequality strict at that date by an expected amount $\delta > 0$, which lowers the bound to $V(h_0) - \beta^t\delta$.

Claim (ii). The objective $\log(wh - a) + b(h)\log a$ is strictly concave and tends to $-\infty$ at both ends of $(0, wh)$, so its unique maximizer is given by $a/(wh - a) = b(h)$, that is, $\lambda(h) = b(h)/(1 + b(h))$. Conversely, a continuous share function λ with values in $(0, 1)$ is generated by the continuous weight $b = \lambda/(1 - \lambda) > 0$.

Claim (iii). Here $T_h = \rho T/h$ and $T_a = \gamma T/a$, so $\chi(h, a) = \rho a/(\gamma h)$ does not depend on ε. With $a = \lambda wh$, $u_a = -1/((1 - \lambda)wh)$ and

$$R(h, a) = \frac{w}{wh - a} + \frac{1}{wh - a} \cdot \frac{\rho a}{\gamma h} = \frac{\gamma + \rho\lambda}{\gamma(1 - \lambda)h}.$$

Substituting into (3), with $T_a(h_t, a_t, \varepsilon_t) = \gamma h_{t+1}/a_t$, gives

$$\frac{1}{(1 - \lambda_t)wh_t} = \beta\,\mathbb{E}_t\left[\frac{\gamma h_{t+1}}{\lambda_t wh_t} \cdot \frac{\gamma + \rho\lambda_{t+1}}{\gamma(1 - \lambda_{t+1})h_{t+1}}\right] = \frac{\beta}{\lambda_t wh_t}\,\mathbb{E}_t\left[\frac{\gamma + \rho\lambda_{t+1}}{1 - \lambda_{t+1}}\right],$$

which rearranges to (5). A constant share satisfies (5) if and only if $\lambda = \beta(\gamma + \rho\lambda)$, that is, $\lambda = \beta\gamma/(1 - \beta\rho)$ or, equivalently, $\beta = \lambda/(\gamma + \rho\lambda)$, consistent with claim (i).

The continuation following Proposition 3. By claim (i), $\Phi^*(h, a) = \beta\,\mathbb{E}[A+B\log(Zh^\rho a^\gamma\varepsilon)] = \beta B\gamma\log a$ plus terms in h alone. Under a technology change, the warm-glow problem $\max_a\{\log(wh - a) + \Phi^*(h, a)\}$ is unchanged, so the share stays at $\lambda_{\mathcal{R}}(\beta)$. By claim (i) applied to the new technology, the altruist's share is $\beta\gamma/(1 - \beta\rho)$ when only Z changes, which is unchanged, and $\beta\tilde{\gamma}/(1 - \beta\rho) \neq \lambda_{\mathcal{R}}(\beta)$ when γ changes to $\tilde{\gamma} \in (0, 1 - \rho)$, $\tilde{\gamma} \neq \gamma$. □

Proof of Proposition 3. Part (a). Let $\tilde{a} = \tilde{\sigma}^{\mathcal{N}}(x)$ be interior. Since $\tilde{a}$ is an interior maximizer of the differentiable function $\tilde{u}(x, \cdot) + \Phi^*(x, \cdot)$ on $\tilde{\Gamma}(x)$,

$$\nabla_a\tilde{u}(x, \tilde{a}) \;=\; -\nabla_a\Phi^*(x, \tilde{a}) \;=\; -\beta\,\nabla_a\mathbb{E}\big[V\big(T(x, \tilde{a}, \varepsilon)\big)\big].$$

By Assumption 1 for the new environment, the altruist's objective $G(a) = \tilde{u}(x, a)+\beta\,\mathbb{E}[\tilde{V}(\tilde{T}(x, a, \varepsilon))]$ is differentiable and strictly concave on $\tilde{\Gamma}(x)$, with unique maximizer $\tilde{\sigma}^{\mathcal{R}}(x)$. If $\tilde{\sigma}^{\mathcal{R}}(x) = \tilde{a}$, then $\tilde{a}$ is an interior maximizer of G, so $\nabla G(\tilde{a}) = 0$. Conversely, if $\nabla G(\tilde{a}) = 0$, then $\tilde{a}$ maximizes G, so $\tilde{\sigma}^{\mathcal{R}}(x) = \tilde{a}$. Hence the two choices coincide if and only if $\nabla G(\tilde{a}) = 0$, that

is,

$$\beta\,\nabla_a \mathbb{E}\big[\tilde{V}\big(\tilde{T}(x,\tilde{a},\varepsilon)\big)\big] \;=\; -\nabla_a \tilde{u}(x,\tilde{a}) \;=\; \beta\,\nabla_a \mathbb{E}\big[V\big(T(x,\tilde{a},\varepsilon)\big)\big].$$

Dividing by $\beta > 0$ gives (6).

Part (b). If $\tilde{u} = u$ and $\tilde{\Gamma} = \Gamma$, the warm-glow problem after the intervention is the pre-intervention problem, because Φ^* is held fixed and does not involve $\tilde{T}$. Hence $\tilde{\sigma}^{\mathcal{N}} = \sigma$, and part (a) applies with $\tilde{a} = \sigma(x)$. □

Proof of Lemma 1. Part (a). This follows from Proposition 1(a) and Proposition 3(a).

Part (b). Since Φ is admissible, σ is continuous, so Φ_c is continuous and continuously differentiable in a. The added term is strictly concave in a for $c > 0$, so $u + \Phi_c$, and after the intervention $\tilde{u} + \Phi_c$, are strictly concave whenever $u + \Phi$ and $\tilde{u} + \Phi$ are. Hence Φ_c is admissible before and after the intervention. It generates σ because, for every $a \in \Gamma(x)$,

$$(u + \Phi_c)(x,a) = (u + \Phi)(x,a) - c\|a - \sigma(x)\|^2 \;\le\; (u + \Phi)(x,\sigma(x)) = (u + \Phi_c)(x,\sigma(x)).$$

Fix a state x at which the counterfactual choice under $\Phi = \Phi_0$ differs from $\sigma(x)$, and take $c \neq c'$. Suppose, to reach a contradiction, that the counterfactual choices under Φ_c and $\Phi_{c'}$ coincide at an interior point a. The two first-order conditions are

$$\nabla_a \tilde{u}(x,a) + \nabla_a \Phi(x,a) - 2c\,(a - \sigma(x)) = 0, \qquad \nabla_a \tilde{u}(x,a) + \nabla_a \Phi(x,a) - 2c'\,(a - \sigma(x)) = 0.$$

Subtracting gives $2(c' - c)(a - \sigma(x)) = 0$, so $a = \sigma(x)$. Then either condition gives $\nabla_a(\tilde{u} + \Phi)(x,\sigma(x)) = 0$. Since $\tilde{u}(x,\cdot) + \Phi(x,\cdot)$ is strictly concave on $\tilde{\Gamma}(x)$, $\sigma(x)$ is its unique maximizer, so the counterfactual choice under Φ is $\sigma(x)$, a contradiction.

The scalar claim in the text. Let $m = 1$, write $g(a) = \tilde{u}_a(x,a) + \Phi_a(x,a)$, which is strictly decreasing by strict concavity, and let $a_0 \neq \sigma(x)$ be the prediction under Φ, with a_0 and $\sigma(x)$ interior, so that $g(a_0) = 0$. Say $a_0 > \sigma(x)$; the other case is symmetric. The function $g(a) - 2c(a - \sigma(x))$ is strictly decreasing, positive at $\sigma(x)$ (where it equals $g(\sigma(x)) > g(a_0) = 0$), and negative at a_0, so its unique zero a_c lies in $(\sigma(x), a_0)$ and is the interior prediction under Φ_c. Moreover, $|a_c - \sigma(x)| = |g(a_c)|/(2c) \le \max_{[\sigma(x),a_0]} |g|/(2c) \to 0$ as $c \to \infty$. □

Proof of Proposition 4. Part (a). The Bellman operator $(\mathcal{T}f)(x) = \max_{a\in\Gamma(x)}\{u(x,a) + \beta\,\mathbb{E}[f(T(x,a,\varepsilon))]\}$ maps bounded continuous functions into bounded continuous functions, by bounded convergence and Berge's theorem, and is a contraction of modulus β in the supremum norm (Stokey et al., 1989). Indeed, for any f, g and $a \in \Gamma(x)$, $u + \beta\,\mathbb{E}[f \circ T] \le u + \beta\,\mathbb{E}[g \circ T] + \beta\|f - g\|_\infty$; maximizing over a and exchanging the roles of f and g gives

$\|\mathcal{T}f - \mathcal{T}g\|_\infty \le \beta\|f-g\|_\infty$. Since $V_k = \mathcal{T}^k\phi$, we have $\|V_{j+1}-V_j\|_\infty \le \beta^j\|V_1-V_0\|_\infty$, and summing gives

$$\|V_k - V\|_\infty \;\le\; \sum_{j=k}^{\infty}\|V_{j+1}-V_j\|_\infty \;\le\; \sum_{j=k}^{\infty}\beta^j\,\|V_1-V_0\|_\infty \;=\; \frac{\beta^k}{1-\beta}\,\|V_1-V_0\|_\infty.$$

Part (b). By Assumption 1(iii), the maximizer $\sigma^{\mathcal{R}}(x)$ is unique, and by Berge's theorem it is continuous. Fix $\epsilon > 0$ and let $S_\epsilon = \{(x,a) : x \in X,\ a \in \Gamma(x),\ \|a - \sigma^{\mathcal{R}}(x)\| \ge \epsilon\}$, a closed subset of the compact graph of Γ. If S_ϵ is empty, there is nothing to prove. Otherwise, the function

$$g(x,a) = \big[u(x,\sigma^{\mathcal{R}}(x)) + \beta\,\mathbb{E}[V(T(x,\sigma^{\mathcal{R}}(x),\varepsilon))]\big] - \big[u(x,a) + \beta\,\mathbb{E}[V(T(x,a,\varepsilon))]\big]$$

is continuous and strictly positive on S_ϵ, so $c(\epsilon) = \min_{S_\epsilon} g > 0$. For any $(x,a) \in S_\epsilon$,

$$\big[u(x,a)+\beta\,\mathbb{E}[V_k(T(x,a,\varepsilon))]\big]-\big[u(x,\sigma^{\mathcal{R}}(x))+\beta\,\mathbb{E}[V_k(T(x,\sigma^{\mathcal{R}}(x),\varepsilon))]\big] \;\le\; -c(\epsilon)+2\beta\,\|V_k-V\|_\infty,$$

which is negative once $2\beta\|V_k - V\|_\infty < c(\epsilon)$. By part (a), this holds for all sufficiently large k, and then every depth-k maximizer satisfies $\|\sigma_k(x) - \sigma^{\mathcal{R}}(x)\| < \epsilon$ for all x.

Part (c). The post-intervention operator $(\tilde{\mathcal{T}}f)(x) = \max_{a\in\tilde{\Gamma}(x)}\{\tilde{u}(x,a) + \beta\,\mathbb{E}[f(\tilde{T}(x,a,\varepsilon))]\}$ is also a contraction of modulus β. Applying parts (a) and (b) to $\tilde{\mathcal{T}}$ with the same terminal value ϕ gives the result. □

Proof of Lemma 2. With $\theta = \mathcal{E} = 0$, resources are $y = (1-\tau)h$, and the environment is that of Example 1 with $w = 1-\tau$, $a = e$, and $\mathbb{E}[\log\varepsilon] = -\sigma_\varepsilon^2/2$.

Shares. Under altruism, Example 1(i) gives the constant share $\lambda_{\mathcal{R}} = \beta\gamma/(1-\beta\rho)$, which is equivalent to the first expression in (16). Warm glow I is Example 1(ii) with $b \equiv \beta_{\mathrm{I}}$. Under warm glow II, $\mathbb{E}[\varepsilon] = 1$ gives $\Phi_{\mathrm{II}}(h,e) = \beta_{\mathrm{II}}\big[\log((1-\tau)Z) + \rho\log h + \gamma\log e\big]$, which is Example 1(ii) with $b \equiv \beta_{\mathrm{II}}\gamma$ plus terms in h alone.

Equivalence. Under (17), $\beta_{\mathrm{I}} = \beta_{\mathrm{II}}\gamma = \beta\gamma/(1-\beta(\rho+\gamma))$, so the three ratios $\lambda_m/(1-\lambda_m)$ coincide, and so do the policy functions. Substituting $e = \lambda(1-\tau)h$ into (10) gives (18). Since the policy and the transition are common, so is the joint distribution of (h_t, e_t) for any initial condition.

Stationary distribution. Equation (18) is a Gaussian AR(1) in $\log h$ with persistence $\rho+\gamma \in (0,1)$ and innovation variance σ_ε^2. Its unique stationary distribution is normal with the mean and variance in (19). In this distribution, $(\log h_t, \log h_{t+1})$ is bivariate normal with equal

variances and correlation $\rho+\gamma$. Since parent and child ranks are both uniformly distributed, the rank-rank slope equals the Spearman correlation, which for a bivariate normal vector with correlation r is $(6/\pi)\arcsin(r/2)$. Because income $(1-\tau)h$ is strictly increasing in h, income ranks coincide with human capital ranks. □

Proof of Proposition 5. Part (a). Fix p and write $\mu = \mu_p^{\mathcal{R}}$, $V = V_p$, and $c(h) = c(h; \sigma_p^{\mathcal{R}}, p)$. At the optimal policy, $V(h) = u(c(h)) + \beta\,\mathbb{E}_\varepsilon[V(h')]$ for every h. Because μ is stationary under the law of motion induced by $\sigma_p^{\mathcal{R}}$, the distribution of h' when $h \sim \mu$ is again μ, so $\int \mathbb{E}_\varepsilon[V(h')]\,d\mu(h) = \int V\,d\mu$, which is finite by hypothesis. Hence $u(c)$ is μ-integrable, and integrating the Bellman equation gives $\int V\,d\mu = \int u(c)\,d\mu + \beta \int V\,d\mu$. Rearranging gives $W(p, \mathcal{R}) = W_{\text{flow}}(p, \mathcal{R})/(1-\beta)$.

Part (b). By definition, $\mathcal{V}_p^{\mathcal{N}}(h) = u\big(c(h; \sigma_p^{\mathcal{N}}, p)\big) + \Phi\big(h, \sigma_p^{\mathcal{N}}(h); p\big)$. Integrating with respect to $\mu_p^{\mathcal{N}}$ gives the result.

Part (c). Write $p_{\mathcal{N}} = p^*(\mathcal{N})$ and $p_{\mathcal{R}} = p^*(\mathcal{R})$, and suppose $p_{\mathcal{N}} < p_{\mathcal{R}}$; the other case is symmetric. By strict single-peakedness, $\text{Reg}(\cdot, m)$ is zero at $p^*(m)$, strictly decreasing below it, and strictly increasing above it. For $p < p_{\mathcal{N}}$, both regrets are strictly larger than at $p_{\mathcal{N}}$, so $\text{MaxReg}(p) > \text{MaxReg}(p_{\mathcal{N}})$. Symmetrically, $\text{MaxReg}(p) > \text{MaxReg}(p_{\mathcal{R}})$ for $p > p_{\mathcal{R}}$. On $[p_{\mathcal{N}}, p_{\mathcal{R}}]$, the function $D(p) = \text{Reg}(p, \mathcal{R}) - \text{Reg}(p, \mathcal{N})$ is continuous and strictly decreasing, with $D(p_{\mathcal{N}}) = \text{Reg}(p_{\mathcal{N}}, \mathcal{R}) > 0$ and $D(p_{\mathcal{R}}) = -\text{Reg}(p_{\mathcal{R}}, \mathcal{N}) < 0$, so it has a unique zero $\hat{p} \in (p_{\mathcal{N}}, p_{\mathcal{R}})$. For $p \in [p_{\mathcal{N}}, \hat{p})$, $\text{MaxReg}(p) = \text{Reg}(p, \mathcal{R}) > \text{Reg}(\hat{p}, \mathcal{R})$, and for $p \in (\hat{p}, p_{\mathcal{R}}]$, $\text{MaxReg}(p) = \text{Reg}(p, \mathcal{N}) > \text{Reg}(\hat{p}, \mathcal{N})$. Hence $\hat{p}$ is the unique minimizer of maximum regret over $\mathcal{P}$, and the two regrets are equal there.

Part (d). Under the approximation, $\text{Reg}(p_{\mathcal{N}}, \mathcal{R}) = \frac{1}{2} a_{\mathcal{R}} \Delta^2$ and $\text{Reg}(p_{\mathcal{R}}, \mathcal{N}) = \frac{1}{2} a_{\mathcal{N}} \Delta^2$, with $\Delta = |p_{\mathcal{R}} - p_{\mathcal{N}}|$, which gives the first ratio. The approximate welfare functions are strictly single-peaked, so part (c) applies to them, and the minimax-regret policy satisfies $a_{\mathcal{R}}(p^{MR} - p_{\mathcal{R}})^2 = a_{\mathcal{N}}(p^{MR} - p_{\mathcal{N}})^2$ with p^{MR} between the two optima. Taking square roots gives the second ratio. The expression for $a_{\mathcal{R}}$ follows by differentiating the identity in part (a) twice, which holds for every p. □

B Additional Quantitative Results

This appendix reports the remaining results of the experiments in Section 4.3. All figures use the calibration in Table 1. Percentiles refer to the baseline stationary distribution, which is common to the three specifications, and are trimmed to $[0.01, 0.99]$.

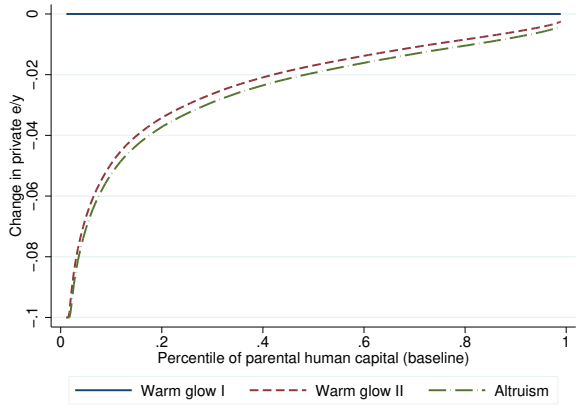
Public education, partial equilibrium. Figure 5 shows responses at $\mathcal{E} = 0.01$. The private education share falls most at the bottom of the distribution, where the public input is large relative to resources, and it falls by more under altruism than under warm glow II at every percentile. The total education share rises most at the bottom under all three specifications. Under altruism, it falls slightly above roughly the 35th percentile, where crowding out exceeds one. The generation alive at the reform gains nothing under warm glow I, because public education changes neither the parent's consumption nor her investment. Under warm glow II and altruism, gains decline with income, and the two curves cross near the 33rd percentile.

Public education, general equilibrium. Figure 6 shows budget-balanced stationary outcomes for $\mathcal{E} \in [0, 0.3]$. Under warm glow I, the tax rate rises least, because the tax base expands with human capital, and private spending is crowded in at every level of $\mathcal{E}$, though by less as $\mathcal{E}$ grows. Under warm glow II and altruism, the stationary crowding-out ratio starts at about 0.97 and 1.18 at $\mathcal{E} = 0.01$ and then falls toward the bound set by baseline private spending, $0.075/\mathcal{E}$, as private spending approaches zero. Average human capital under altruism falls slightly for the smallest expansions and rises thereafter, and for large $\mathcal{E}$ the two specifications become nearly indistinguishable. Welfare is hump-shaped under all three specifications. In utility units, the altruistic gain is the largest for all $\mathcal{E} \geq 0.02$. In consumption equivalents, the warm-glow-I gain is the largest up to about $\mathcal{E} = 0.09$ and the warm-glow-II gain is the largest beyond it, while the altruistic gain overtakes the warm-glow-I gain near $\mathcal{E} = 0.17$.

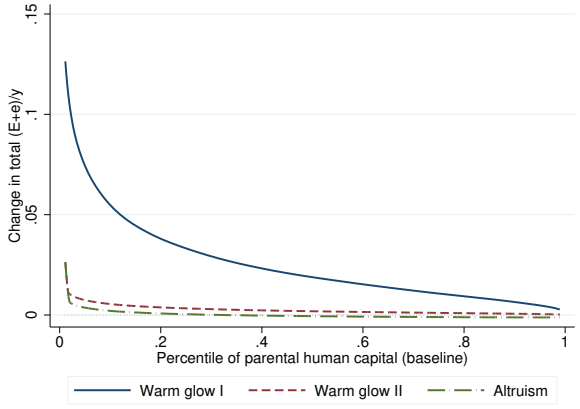
Universal basic income. Figure 7 shows PE welfare gains by percentile at $\theta = 0.05$ and GE outcomes for $\theta \in [0, 0.3]$. PE gains are very large at the bottom, where the transfer is large relative to resources and utility is logarithmic. They are flatter under altruism, falling from about 34 percent at the bottom to 8 percent at the top, than under warm glow II, where they fall from about 93 to 5 percent. The altruistic parent values the transfers her descendants will receive, and because intergenerational persistence is only 0.36, their expected position is closer to the mean than hers. In GE, the tax rate rises most under altruism, because human capital, and with it the tax base, falls most. Welfare rises throughout the

range under warm glow I and is hump-shaped under warm glow II and altruism. In utility units, the altruistic gain is the largest throughout. In consumption equivalents, it is the smallest until it overtakes the warm-glow-II gain at about $\theta = 0.24$.

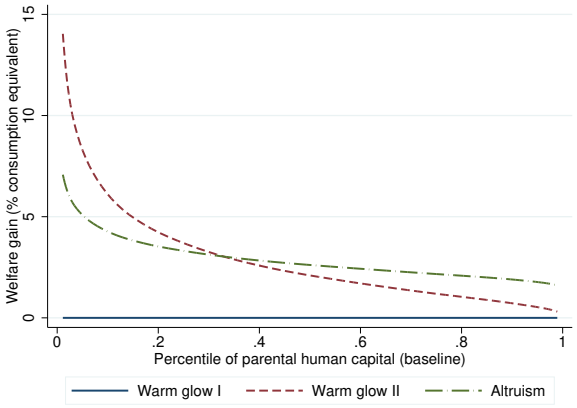
Transition dynamics. Figure 8 traces mean human capital and the welfare of each generation in PE, starting from the baseline stationary distribution. Generation 1 is alive at the reform, and its welfare is evaluated on the baseline distribution with the post-reform value function. Human capital starts to change in generation 2, and the paths converge within about four generations. Altruistic welfare jumps in generation 1 and stays flat, because the continuation value capitalizes all future gains at the time of the reform. Warm-glow welfare instead rises gradually as the distribution of human capital shifts. Under public education, warm-glow-I welfare is unchanged in generation 1 and rises to 4.7 percent as human capital accumulates.



(a) Change in private e/y



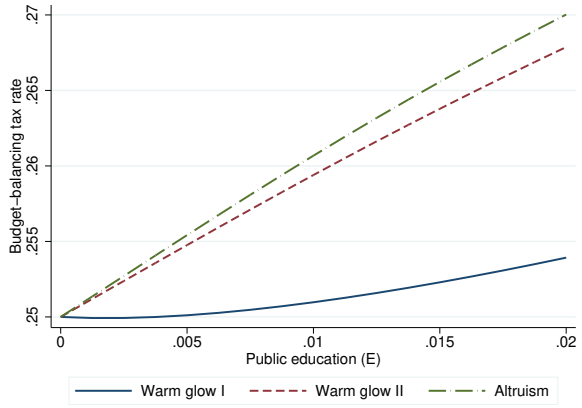
(b) Change in total $(\mathcal{E} + e)/y$



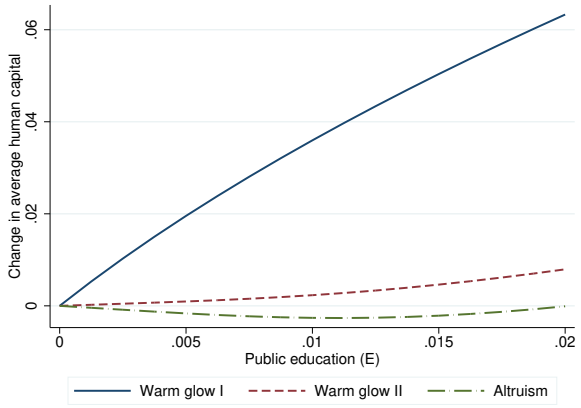
(c) Welfare gain, % consumption equivalent

Figure 5: Public education, partial equilibrium ($\mathcal{E} = 0.01$), by baseline percentile

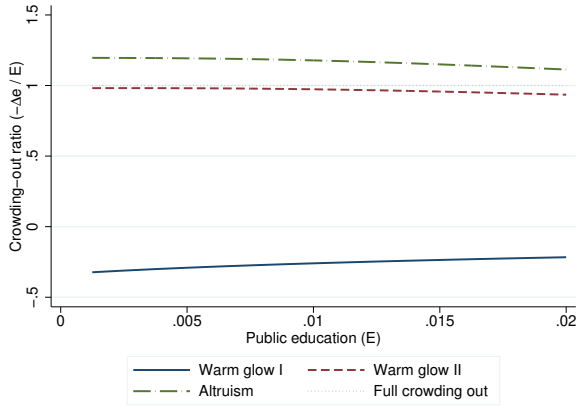
Note: Post-intervention policy and value functions at baseline $\tau = 0.25$, evaluated at each percentile of the baseline distribution. In panel (b), the dotted line marks zero.



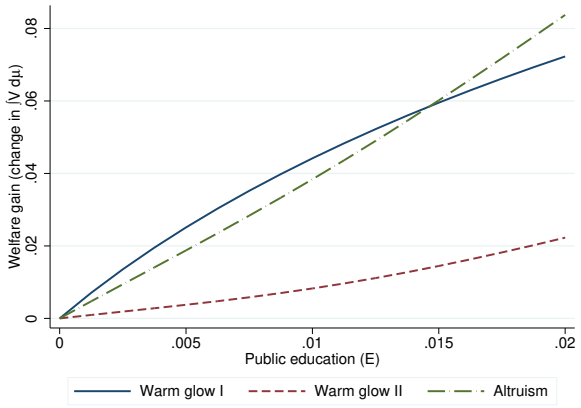
(a) Budget-balancing tax rate



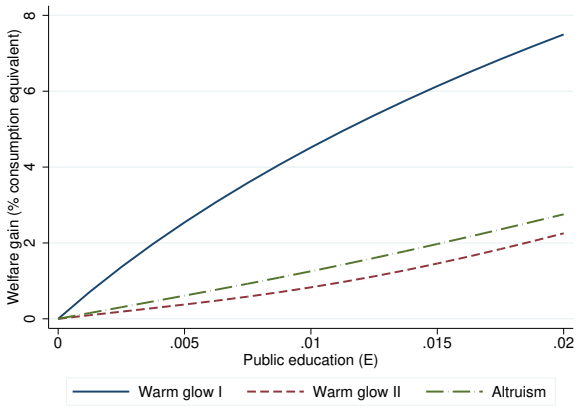
(b) Change in mean human capital



(c) Crowding out, stationary means



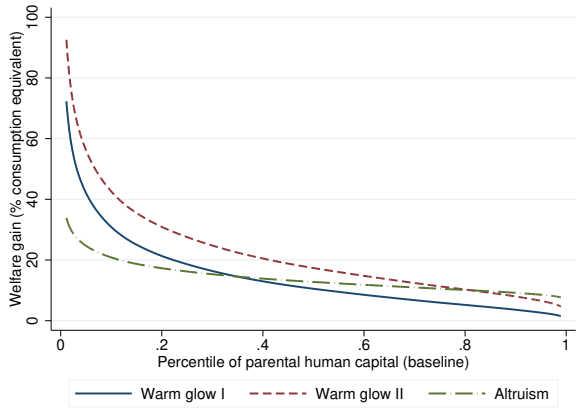
(d) Welfare gain, utils



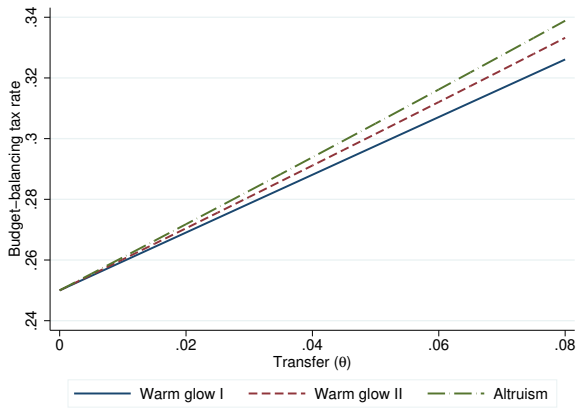
(e) Welfare gain, % consumption equivalent

Figure 6: Public education, general equilibrium, $\mathcal{E} \in [0, 0.3]$

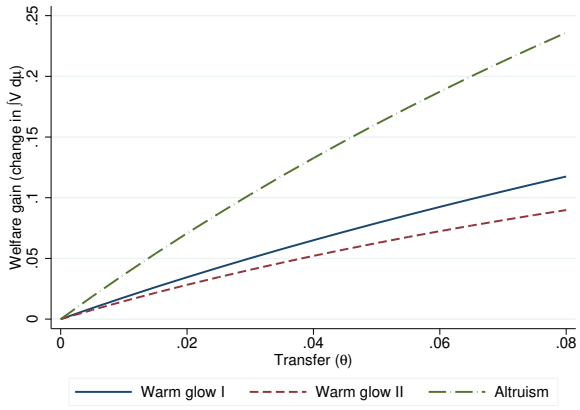
Note: Budget-balanced stationary equilibria on a grid with step 0.01. Welfare is $W(\mathcal{E}, m) - W(0, m)$ from (28), converted to consumption equivalents with (26) in panel (e).



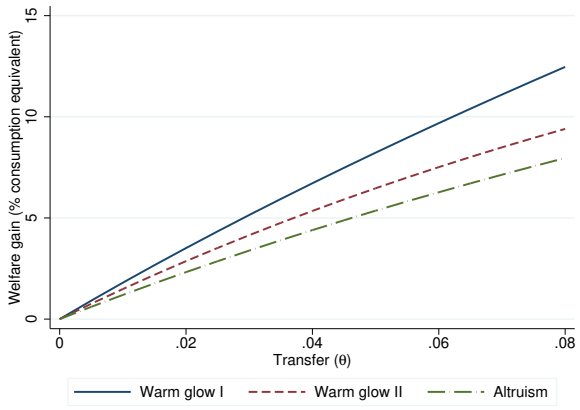
(a) PE welfare gain by percentile, $\theta = 0.05$



(b) GE budget-balancing tax rate



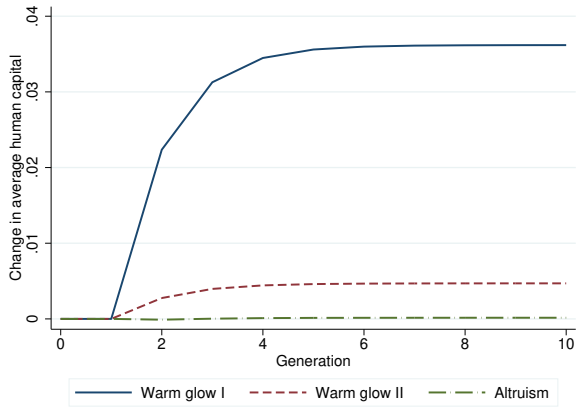
(c) GE welfare gain, utils



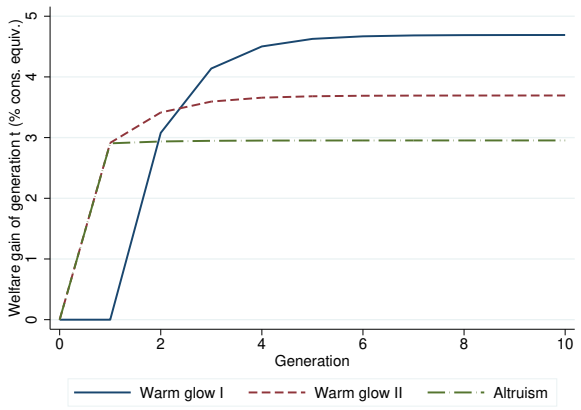
(d) GE welfare gain, % consumption equivalent

Figure 7: Universal basic income: welfare and fiscal outcomes

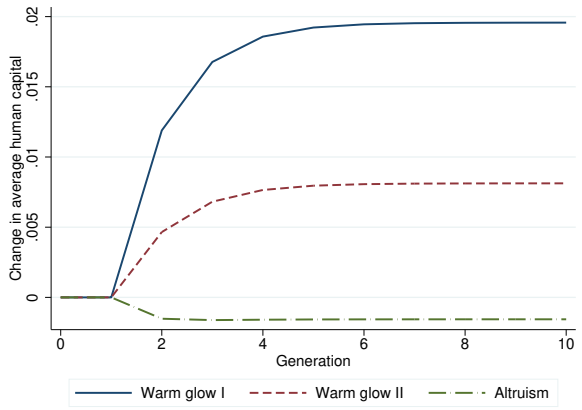
Note: Panel (a) evaluates post-intervention value functions at baseline τ on the baseline distribution. Panels (b) to (d) report budget-balanced stationary equilibria on a grid with step 0.01 for $\theta \in [0, 0.3]$.



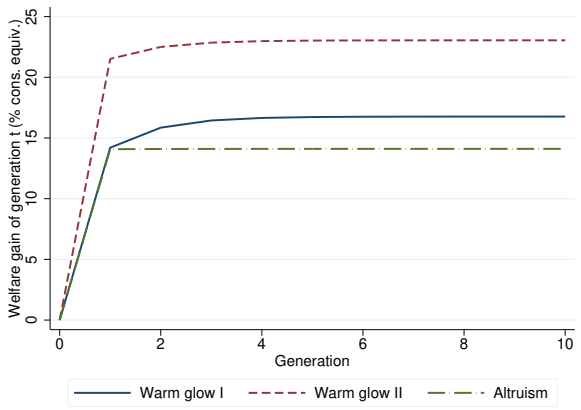
(a) Public education: mean human capital



(b) Public education: welfare of generation t



(c) UBI: mean human capital



(d) UBI: welfare of generation t

Figure 8: Partial-equilibrium transition dynamics ($\mathcal{E} = 0.01$, $\theta = 0.05$)

Note: Generation 0 is the baseline; generation 1 is alive at the reform and starts from the baseline distribution. Welfare of generation t is $\int \mathcal{V}_p^m \, d\mu_t^m$ relative to baseline welfare, in consumption equivalents.

C Numerical Implementation and Accuracy

Discretization. Human capital takes 201 values spaced uniformly in $\log h$ on $[0.005, 80]$. The shock $\log \varepsilon$ is discretized with 15 Gauss-Hermite nodes, rescaled so that $\mathbb{E}[\varepsilon] = 1$ exactly. The child's human capital $h' = \bar{h}\,\varepsilon$ is assigned to the two adjacent grid points by a lottery that is linear in h. The mass of the stationary distribution at either end of the grid is below 10^{-8} at baseline and in the PE experiments, and below 10^{-6} in every GE experiment.

Solution. We solve the altruistic problem by value function iteration to a supremum-norm tolerance of 10^{-9}, interpolating continuation values linearly in $\log h$ and choosing $e \in [0, y)$ by bounded one-dimensional optimization, which allows the corner $e = 0$. Warm glow I, and warm glow II with $\theta = 0$, have closed-form solutions; warm glow II with $\theta > 0$ is solved numerically. Stationary distributions are computed by iterating on the transition until the L^1 change falls below 10^{-12}. In GE, the tax rate is updated as $\tau = (G + \theta + \mathcal{E})/\int h\, d\mu$ until it changes by less than 10^{-9}, and budget gaps are of order 10^{-9}. The depth hierarchy applies the post-intervention Bellman operator to ϕ up to $k = 20$. Supremum norms are taken over grid points whose baseline cumulative probability lies in $[0.001, 0.999]$, and L^1 norms are weighted by the baseline distribution.

Policy grids, optima, and minimax regret. GE outcomes are computed on grids with step 0.01 on $[0, 0.3]$ for each instrument, 31 points each, and the PE levels $\mathcal{E} = 0.01$ and $\theta = 0.05$ are grid points. Each model's optimum is located by golden-section search, to a tolerance of 10^{-4}, between the neighbors of its best grid point, solving the budget-balanced stationary equilibrium at every trial policy. The curvature a_m is a central second difference with step 0.01. Minimax-regret policies are located by golden-section search on maximum regret between the neighbors of the best grid point, solving all models at every trial policy. Regret in Figure 4 is measured relative to these optima.

Calibration. The closed form of Lemma 2 provides the starting point. For the altruistic specification, we then choose $(Z, \gamma, \sigma_\varepsilon, \beta)$ by Nelder-Mead to minimize the sum of squared percentage deviations of the four grid moments from their targets, computing the rank-rank slope as the regression coefficient of child rank on parent rank. The final loss is below 10^{-15}. The warm-glow weights are then backed out from λ using (16), and they match (17) to seven digits. As a check, calibrating each warm-glow specification separately, with its own technology parameters, recovers the same $(Z, \gamma, \sigma_\varepsilon)$ to within 5×10^{-9} and the weights implied by (17).

Accuracy checks. At the common baseline, $\max|e/y - \lambda|$ over the support is 10^{-8} under altruism and 3×10^{-17} under warm glow, the L^1 distance between the altruistic and warm-glow stationary distributions is 4×10^{-8}, and the slope of the altruistic value function in $\log h$ equals B to within 2×10^{-14}. The identity of Proposition 5(a) holds to within 0.03 percent: $W = -2.27815$, against $W_{\text{flow}}/(1-\beta) = -2.27869$. The small gap reflects the difference between interpolated continuation values and the lottery transition. At $p = 0$, every GE computation reproduces baseline welfare to within 10^{-10}.